

Computation Graphs, Backpropagation and Autodiff

How gradients flow through neural networks

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ES 667 · Deep Learning · IIT Gandhinagar

Backprop vocabulary

Lecture 2 built the forward map; today we reverse it

Last time, an MLP turned an input into a prediction and then one scalar loss. Today we reuse that **same graph** in reverse:

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General pattern

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Backward: compute sensitivities loss
sensitivity for every parameter θ_j

Concrete scalar example

Forward: with $(w, x, b, y) = (2, 3, 1, 10)$

$$\hat{y} = 2 \cdot 3 + 1 = 7, \quad \mathcal{L} = (7 - 10)^2 = 9$$

Backward:

$$\left(\frac{\partial \mathcal{L}}{\partial w}, \frac{\partial \mathcal{L}}{\partial b} \right) = (-18, -6)$$

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same executed graph · values flow forward, gradients flow backward

Read the notation before the algorithm

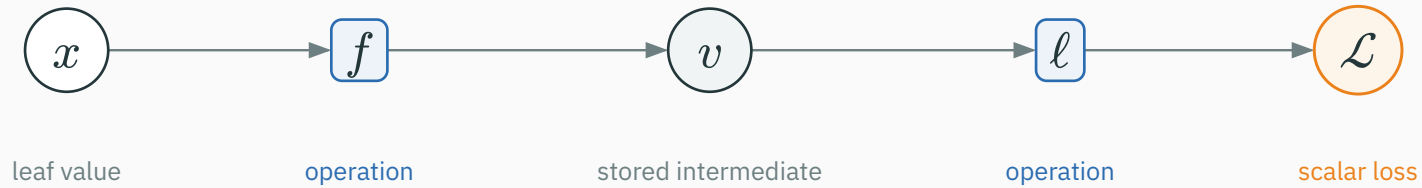
Symbol	Meaning in this lecture
u, v, z	scalars
$\mathbf{x}, \mathbf{z}, \boldsymbol{\theta}$	vectors (bold lowercase)
$\mathbf{W}, \mathbf{X}$	matrices (bold uppercase)
$\partial \mathcal{L} / \partial u$	how the loss changes when scalar u changes
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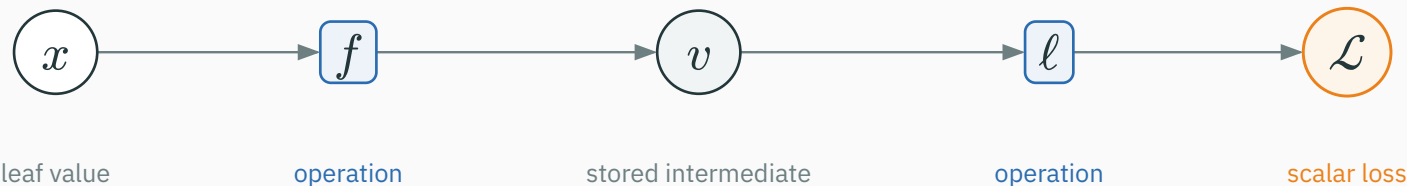
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We keep the full $\partial\mathcal{L}/\partial(\cdot)$ notation visible throughout: the numerator is always the final loss; the denominator names the stored value whose derivative we want.

Our computation-graph convention

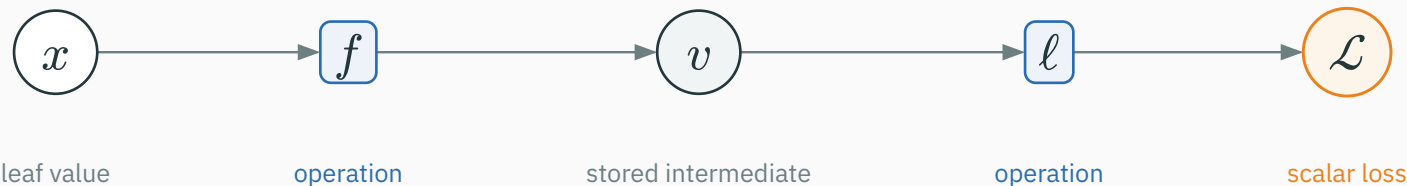


Our computation-graph convention



white circle	given value: input, parameter, or target — a graph leaf
gray circle	computed intermediate value, saved for backward when needed
blue box	operation: consumes value(s) and produces a value
gray arrow	forward dependency from a value to the operation that uses it
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circles store values • boxes transform them • arrows show dependency

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formula · approximate value · reverse sweep through the executed graph

Reverse-mode AD computes the derivative from local rules, up to floating-point arithmetic. Central difference independently approximates the same value: $g_{\text{FD}} = \frac{\mathcal{L}(\theta_j + \varepsilon) - \mathcal{L}(\theta_j - \varepsilon)}{2\varepsilon}$.

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import torch
def loss(w): return (w*3.0 + 1.0 - 10.0)**2
w, eps = 2.0, 1e-5
g_fd = (loss(w + eps) - loss(w - eps)) / (2*eps)
wt = torch.tensor(w, dtype=torch.float64, requires_grad=True)
loss(wt).backward()
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Use float64 and check only a few coordinates. Finite differences debug gradients; they do not train models because every checked coordinate needs two extra forward passes.

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Symbolic to reason • finite differences to check • reverse mode to train.

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We need all P derivatives $\partial \mathcal{L} / \partial \theta_j$. Backprop computes them together in **one reverse sweep**, without perturbing parameters one at a time.

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1 · local rule	How does one operation pass a gradient backward?

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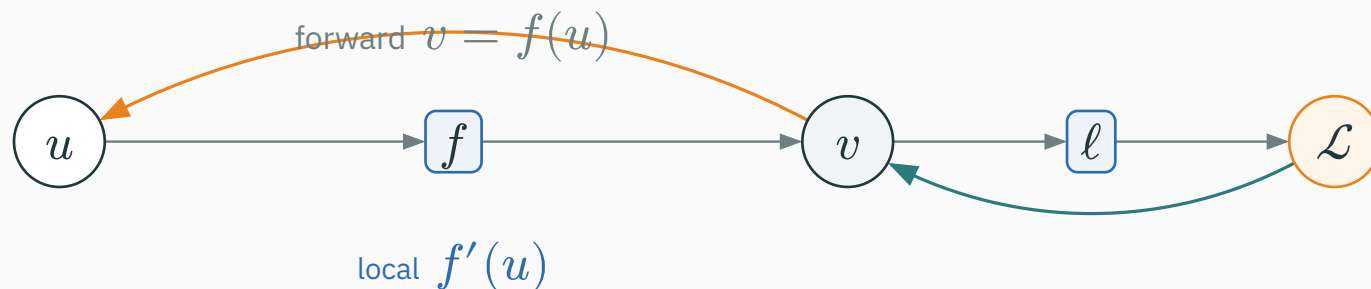
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5 · depth + gradient flow	What happens when many local gradients multiply?

One local operation

Zoom in on one operation. The value circle u enters the boxed operation f ; its output is stored in the value circle v .

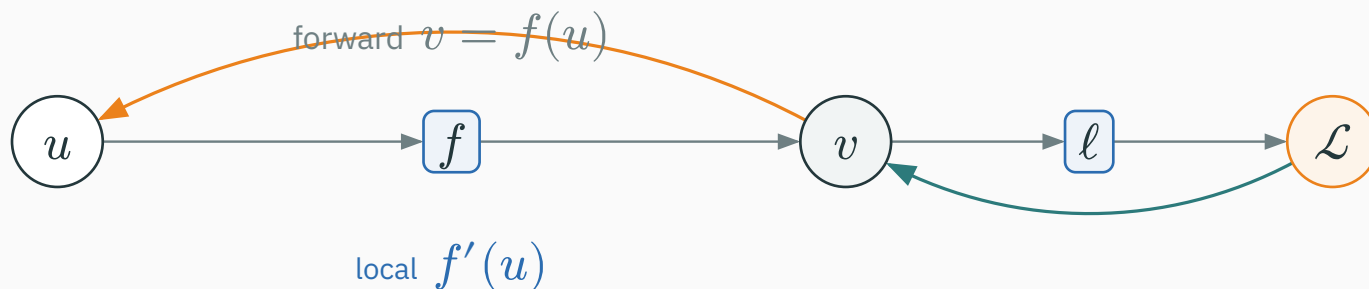
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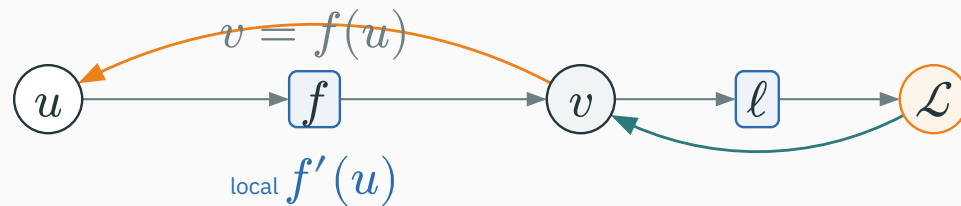


backward computes $\frac{\partial \mathcal{L}}{\partial u} = \frac{\partial \mathcal{L}}{\partial v} \cdot f'(u)$

Every operation multiplies the arriving gradient by its local gradient

D DERIVATION

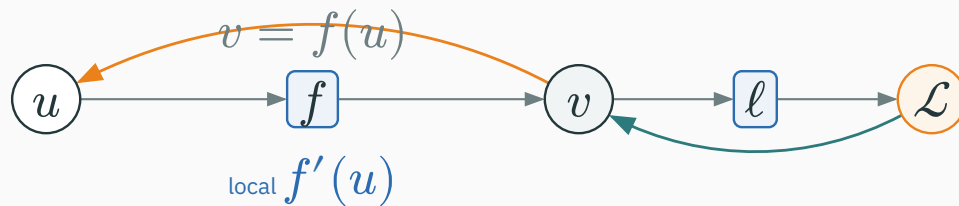
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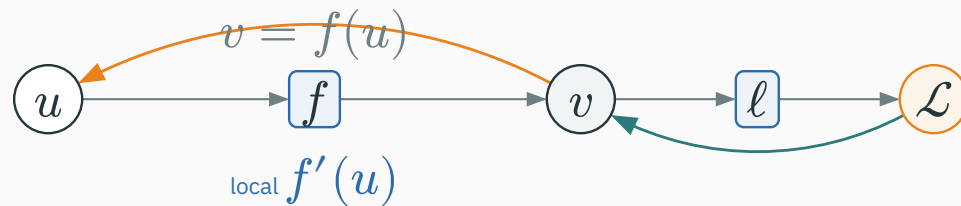


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Downstream gradient

sent back to u

$$\frac{\partial \mathcal{L}}{\partial u}$$

Upstream gradient

arriving at v

$$\frac{\partial \mathcal{L}}{\partial v}$$

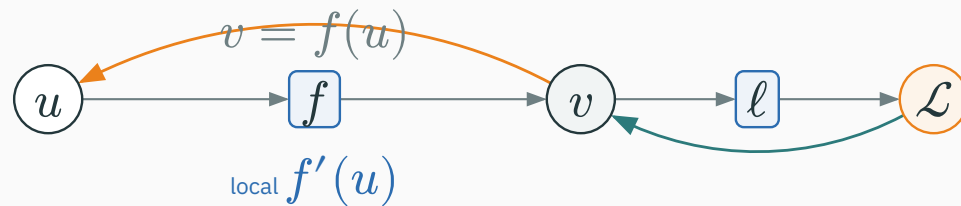
Local gradient

this operation's slope

$$f'(u) = \frac{\partial v}{\partial u}$$

The chain rule is one reusable local rule

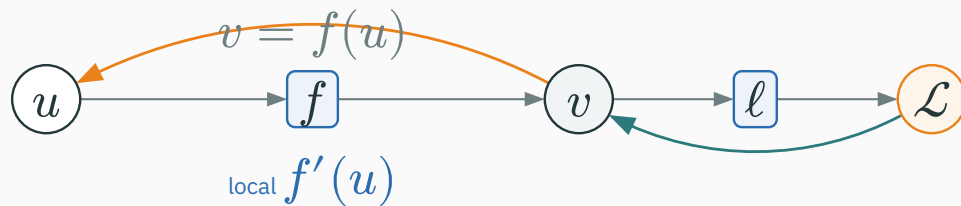
For every scalar operation $v = f(u)$:



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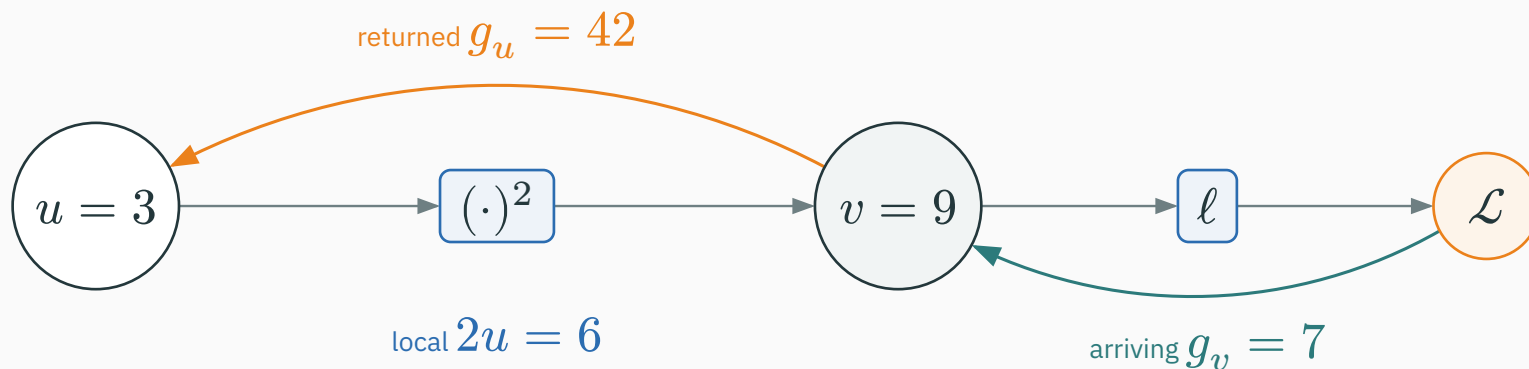
The operation multiplies its **upstream gradient** by its **local gradient** to produce the **downstream gradient**. It needs no global formula for the whole network.

Example: a square operation

For $v = u^2$ at $u = 3$, the forward value is $v = 9$ and the **local slope** is $\partial v / \partial u = 2u = 6$. Suppose the rest of the graph sends $g_v = \partial \mathcal{L} / \partial v = 7$ back to this square.

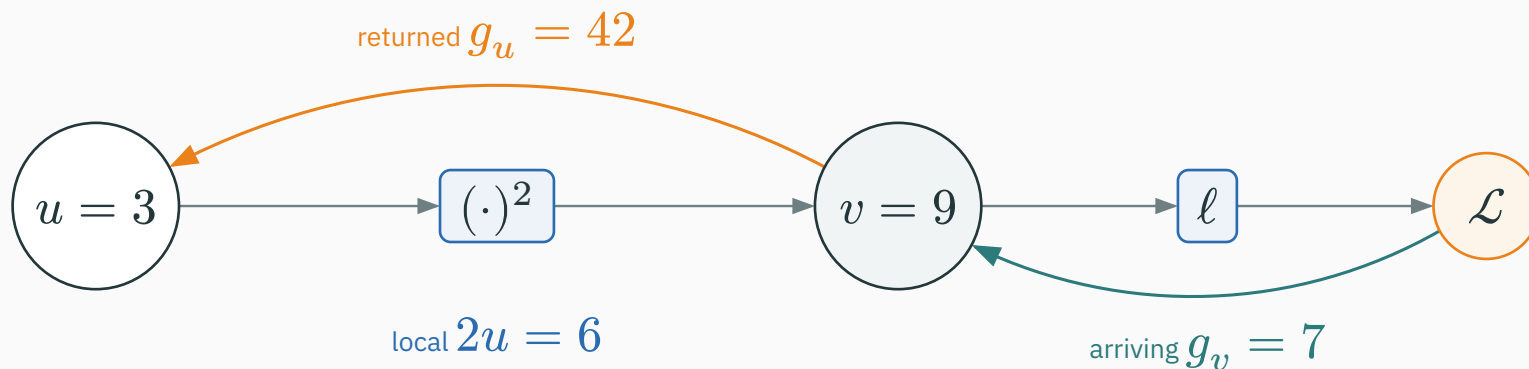
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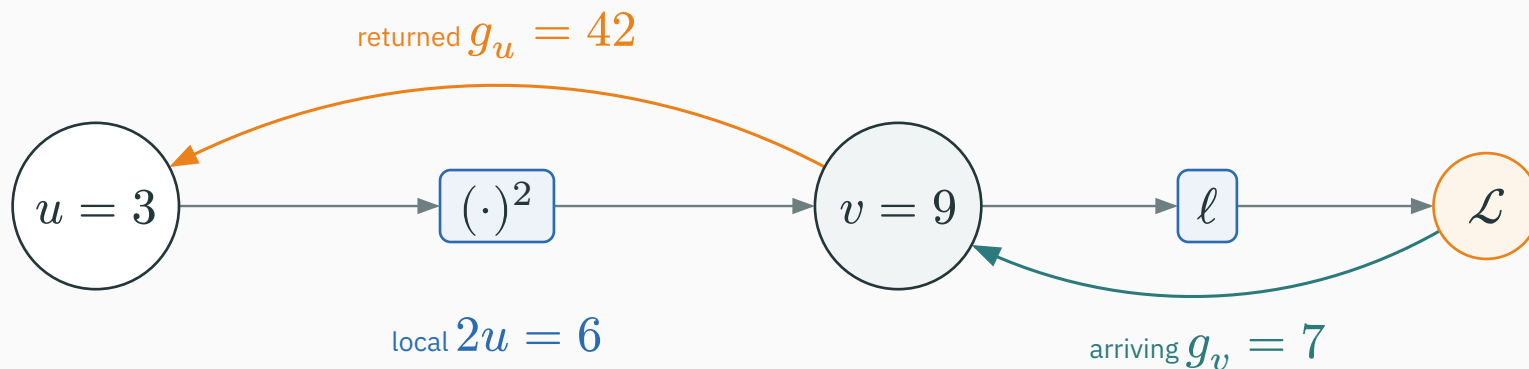
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$$\frac{\partial \mathcal{L}}{\partial u} = \frac{\partial \mathcal{L}}{\partial v} \cdot 2u = 7 \cdot 6 = 42.$$

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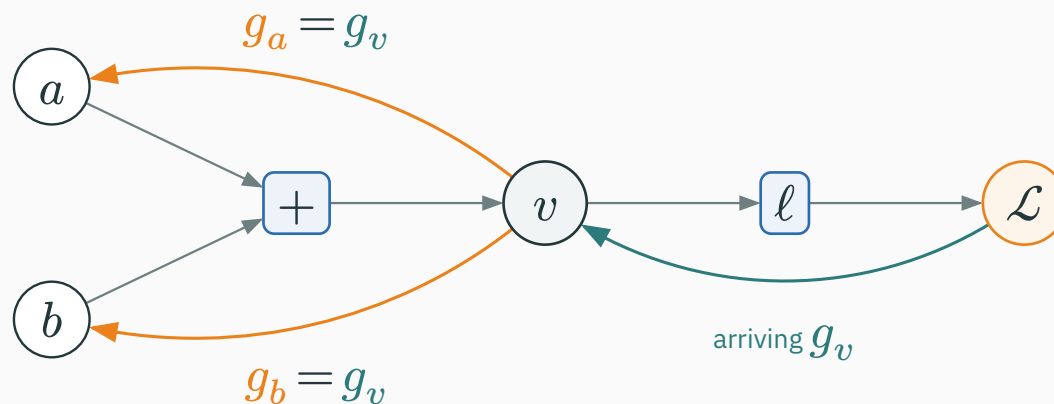
The square operation multiplies the arriving $\partial \mathcal{L} / \partial v = 7$ by its **local gradient** $2u = 6$ to produce $\partial \mathcal{L} / \partial u = 42$.

Addition copies the arriving gradient

Operation $v = a + b$. Let $g_v = \partial \mathcal{L} / \partial v$ arrive. Local derivatives: $\partial v / \partial a = 1$, $\partial v / \partial b = 1$.

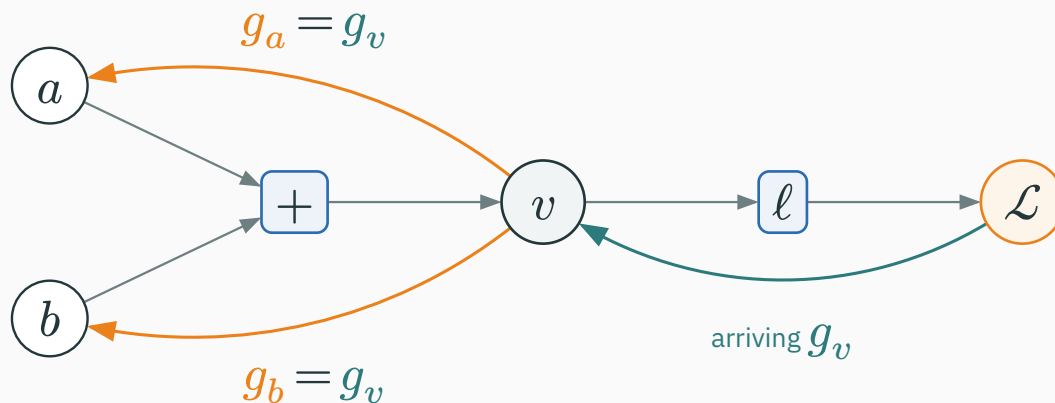
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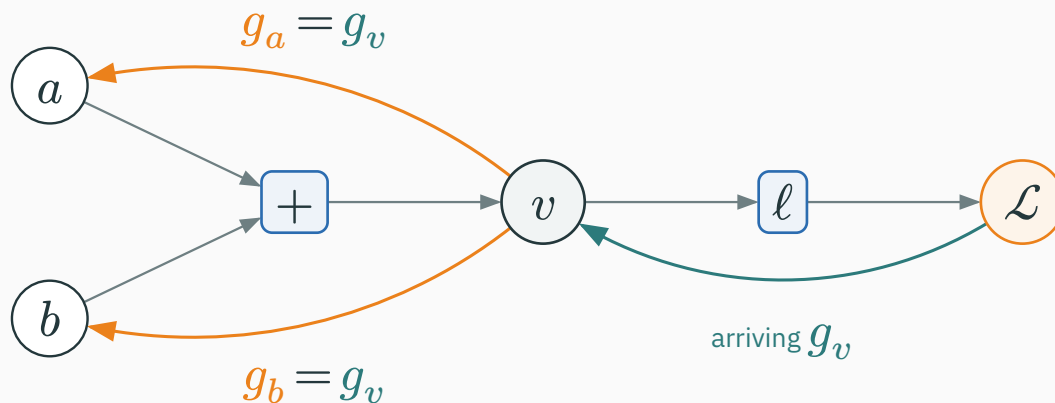
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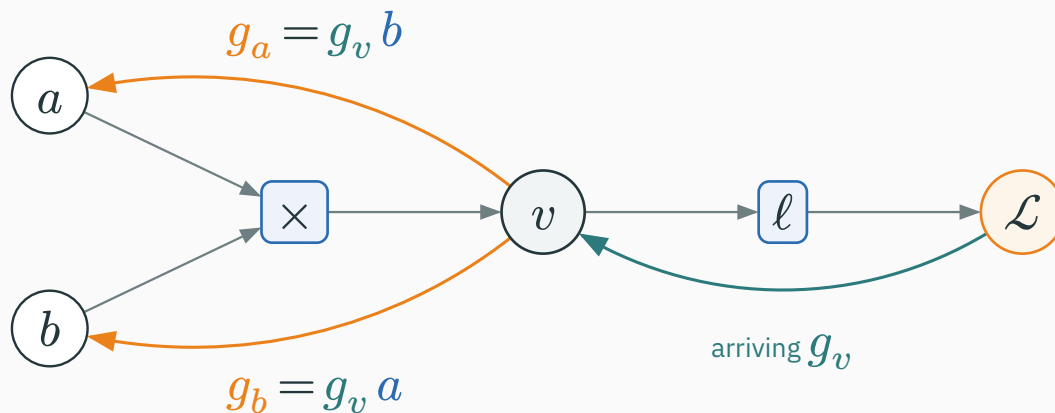
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+ copies the arriving derivative into a downstream derivative for every input.

Operation $v = ab$. Let $g_v = \partial \mathcal{L} / \partial v$ arrive. Local derivatives: $\partial v / \partial a = b$, $\partial v / \partial b = a$.

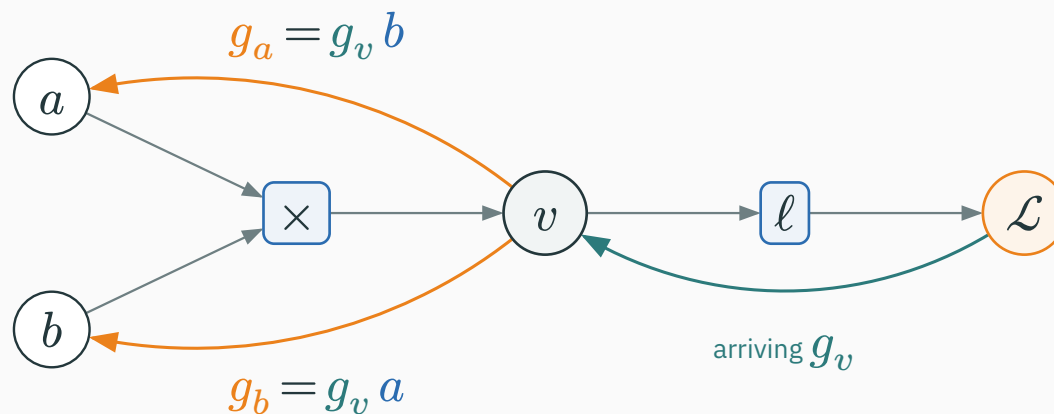
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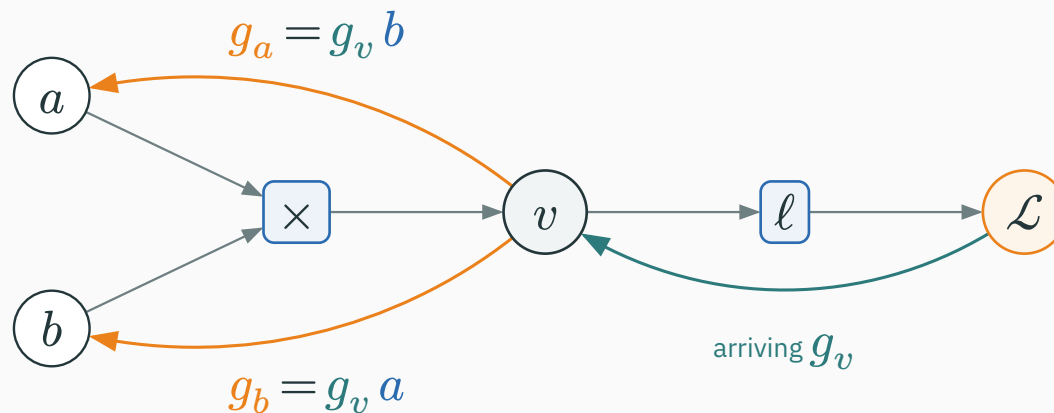
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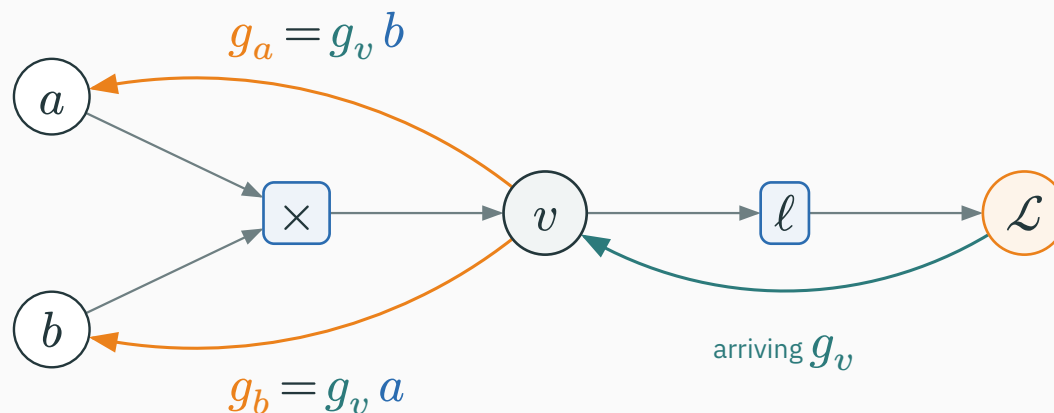


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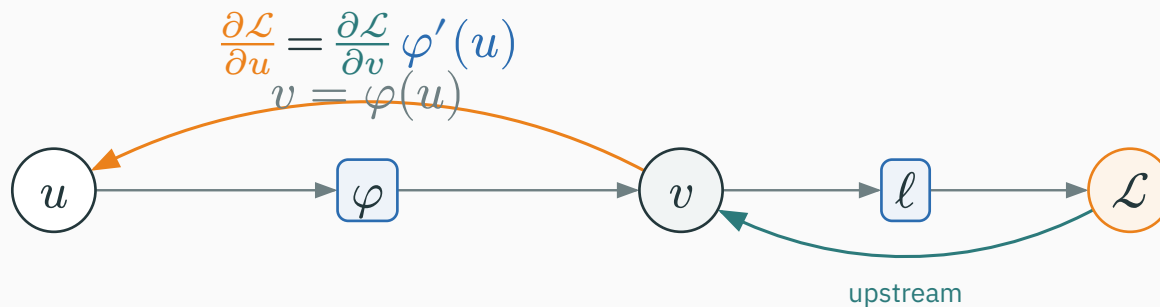
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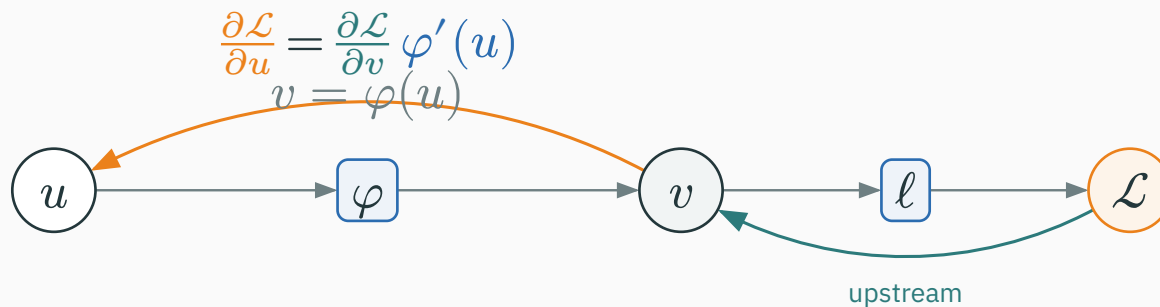
| If $a = 2$, $b = 5$, and $\partial \mathcal{L} / \partial v = 3$, then $\partial \mathcal{L} / \partial a = 15$ and $\partial \mathcal{L} / \partial b = 6$.

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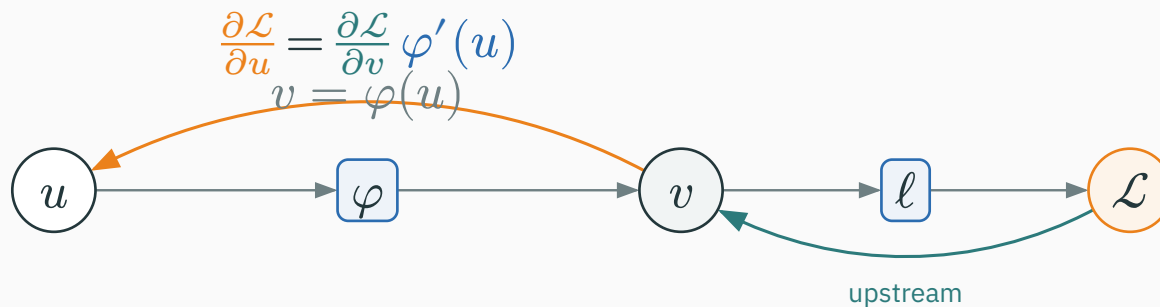


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| **sigmoid** σ : $\varphi'(u) = \sigma(u)(1 - \sigma(u))$

| **ReLU**: $\varphi'(u) = 1$ for $u > 0$, and 0 for $u < 0$

Pop quiz: what crosses an inactive ReLU?

QUESTION

Let $v = \text{ReLU}(u)$. If $u = -2$ and the upstream gradient is $\partial \mathcal{L} / \partial v = 5$, what downstream gradient $\partial \mathcal{L} / \partial u$ is sent back?

A. 5

B. -10

C. 0

D. undefined because $u < 0$

Answer: an inactive ReLU blocks the gradient

A ANSWER

Correct: C — 0

Why: For $u < 0$, the local gradient is $\partial v / \partial u = 0$. Therefore downstream = upstream $\times$ local = $5 \times 0 = 0$.

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Backward: $5 \times 0 \rightarrow \partial \mathcal{L} / \partial u = 0$

At exactly $u = 0$, ReLU's ordinary derivative is undefined. Autodiff software must choose a convention; commonly it returns 0.

The rule table — memorize this

Forward operation	Contribution sent backward
$v = a + b$	$\frac{\partial \mathcal{L}}{\partial a} += \frac{\partial \mathcal{L}}{\partial v} \cdot 1$ $\frac{\partial \mathcal{L}}{\partial b} += \frac{\partial \mathcal{L}}{\partial v} \cdot 1$
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$v = \varphi(u)$	$\frac{\partial \mathcal{L}}{\partial u} += \frac{\partial \mathcal{L}}{\partial v} \cdot \varphi'(u)$

Use +=, not = — a value consumed by several operations accumulates gradient from **all** of them.

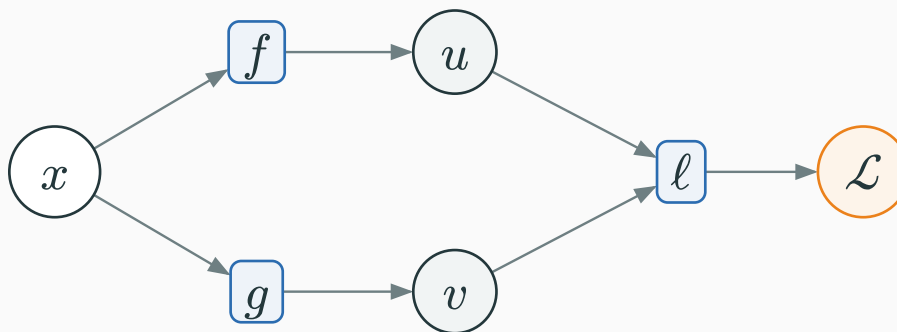
Gradients accumulate at branches

Why gradients add at a branch

Imagine nudging x from 4 to 4.01. The **same nudge** travels down both routes to the loss:

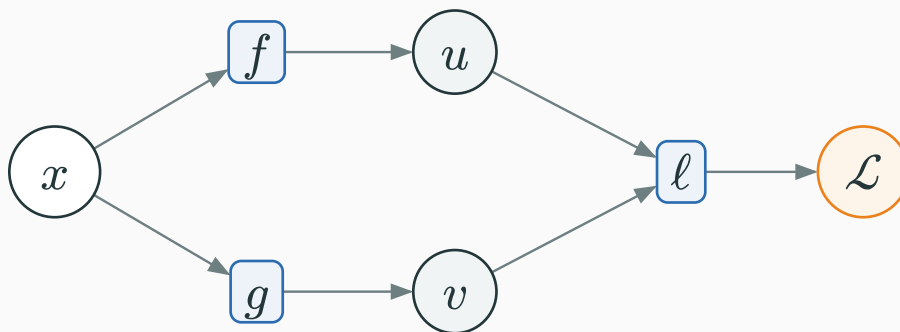
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Via $u = x^2$: the local slope is $2x = 8$, so $\Delta x = 0.01$ changes the loss by about $8 \cdot 0.01 = 0.08$.

Via $v = 3x$: the local slope is 3, so the same nudge changes the loss by about $3 \cdot 0.01 = 0.03$.

$$\text{total slope} = \text{total loss change} \div \text{nudge} = \frac{0.08+0.03}{0.01} = 8 + 3 = 11$$

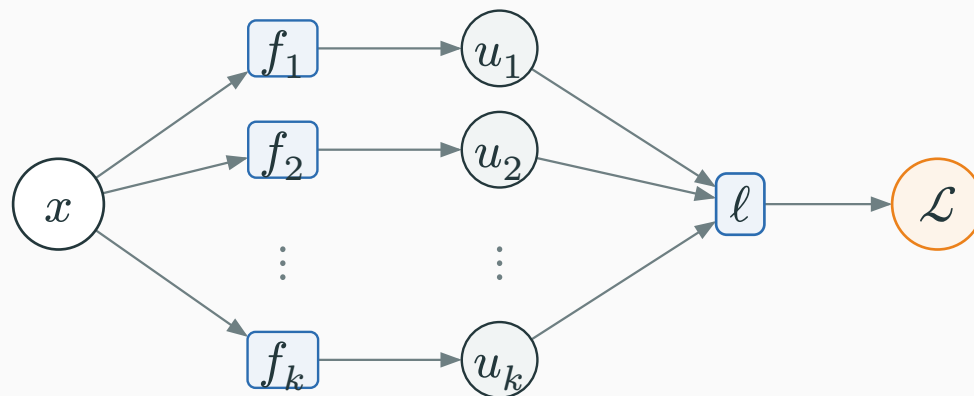
The multivariable chain rule adds every path

D DERIVATION

The nudge experiment generalizes. If x influences $\mathcal{L}$ through $u_1, \dots, u_k$:

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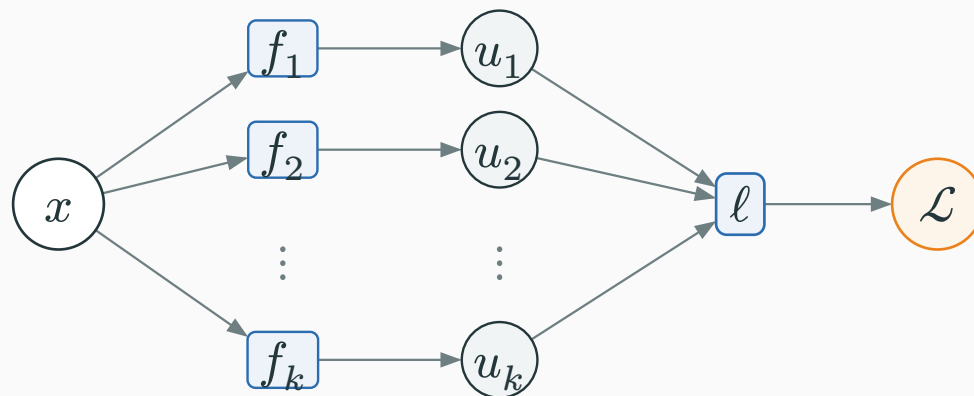
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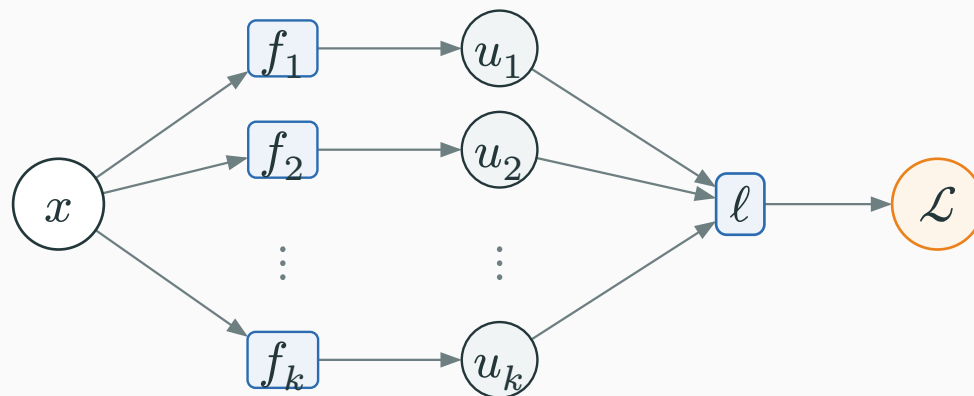


branch i contributes upstream $\partial \mathcal{L} / \partial u_i \times$ local $\partial u_i / \partial x$

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each path contributes one chain-rule product; the products add

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Now let us get the **same** answer by backprop.

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$$\frac{\partial\mathcal{L}}{\partial x} = 8 + 3 = 11. \quad \checkmark$$

PyTorch adds the same two paths

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import torch

x = torch.tensor(4.0, requires_grad=True)
u = x**2           # path 1
v = 3*x           # path 2
L = u + v
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| **via u :** $\partial_{\frac{u}{\partial}} x = 2x = 8$

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autograd accumulates both downstream contributions: $8 + 3 = 11$

The central worked example

Predict $\hat{y} = wx + b$, squared-error loss $\mathcal{L} = (\hat{y} - y)^2$.

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Compute $\partial\mathcal{L}/\partial w$, $\partial\mathcal{L}/\partial x$, $\partial\mathcal{L}/\partial b$, and $\partial\mathcal{L}/\partial y$ by backpropagation; verify the four values with PyTorch.

Break into primitives

Decompose the loss into elementary operations and name each stored result:

$$m = wx, \quad a = m + b, \quad e = a - y, \quad \mathcal{L} = e^2.$$

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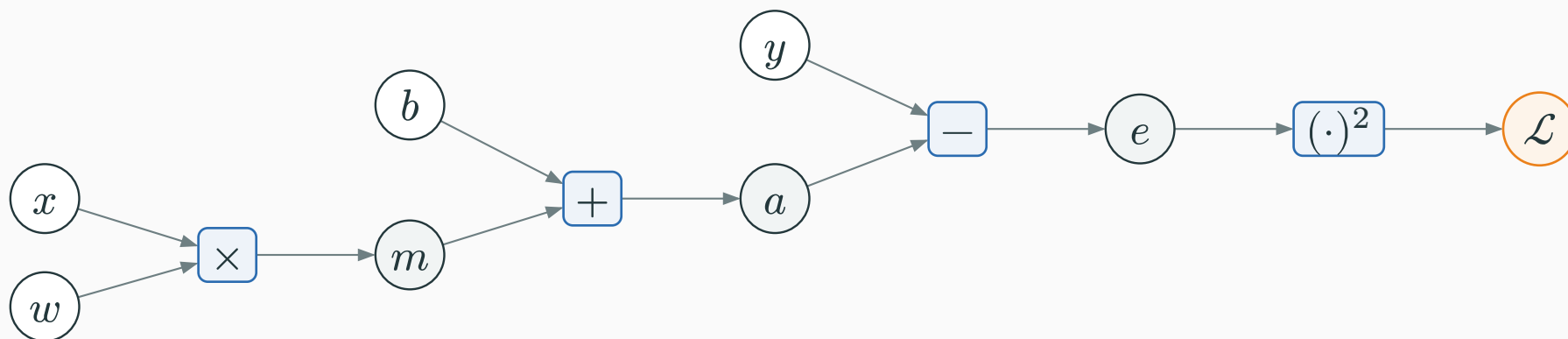
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value m after $\times$ • value a after $+$ • value e after $-$ • value $\mathcal{L}$ after $(\cdot)^2$

The symbols m , a , and e name stored intermediate values. They are circles in the graph; the four operations are boxes.

The computation graph



Given values	w, x, b, y	Stored values	m, a, e
Operations	$\times, +, -, (\cdot)^2$	Scalar loss	$\mathcal{L}$

Stored value	Operation that produces it	Numeric value
m	$w \times x$	$2 \cdot 3 = 6$
a	$m + b$	$6 + 1 = 7$
e	$a - y$	$7 - 10 = -3$
$\mathcal{L}$	e^2	$(-3)^2 = 9$

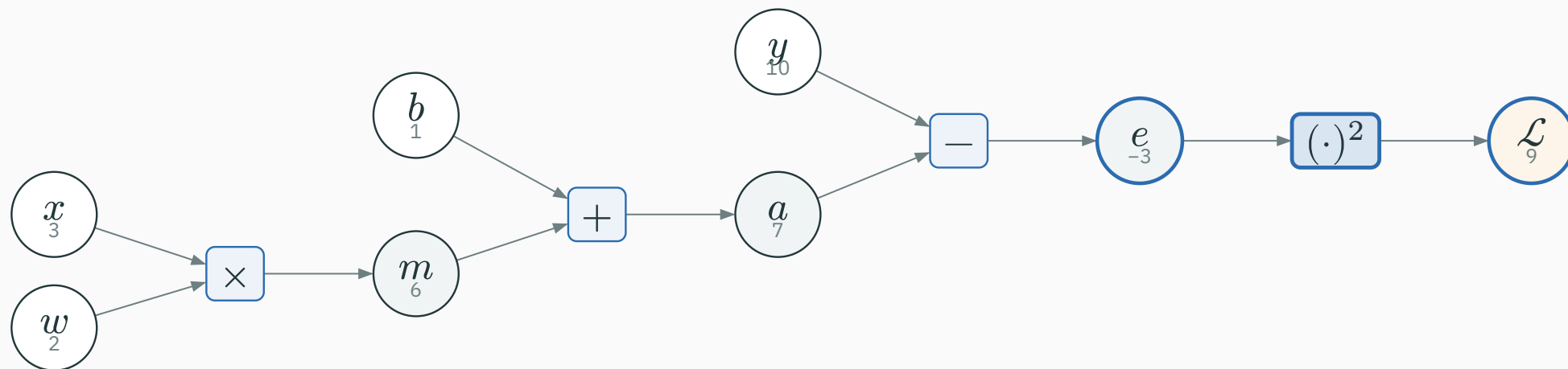
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$\mathcal{L} = 9$ — now run the graph **backward**

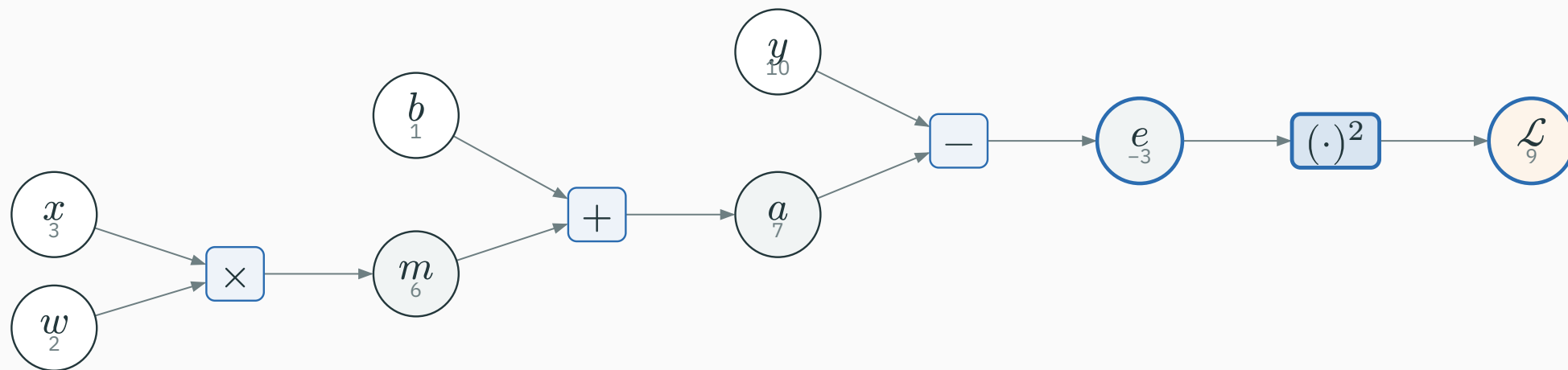
Seed the output: $\partial \mathcal{L} / \partial \mathcal{L} = 1$.

Backward: seed and the square operation

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Square operation $\mathcal{L} = e^2$, local $\partial \mathcal{L} / \partial e = 2e$:

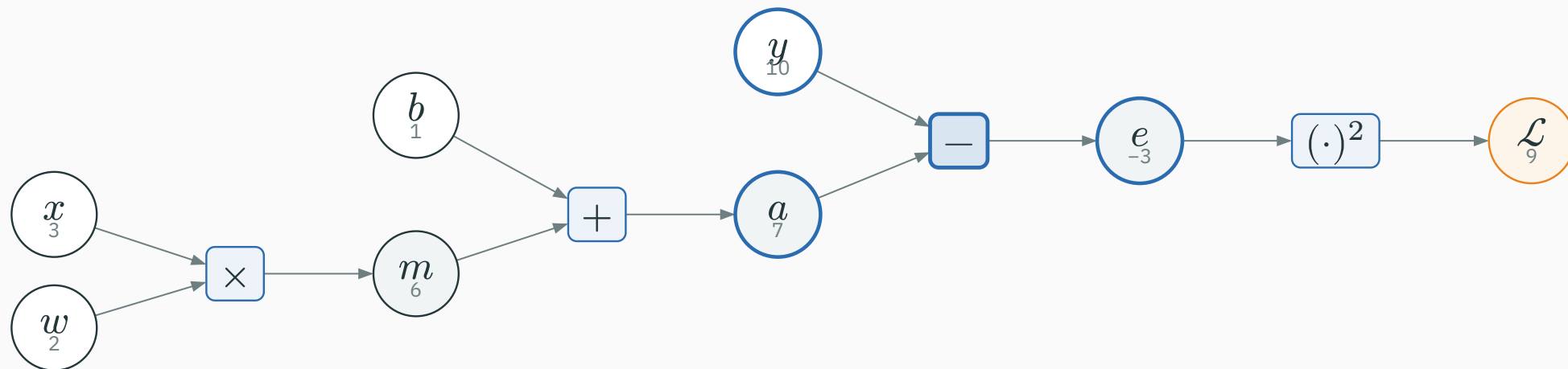
$$\frac{\partial \mathcal{L}}{\partial e} = 1 \cdot 2e = 1 \cdot 2(-3) = -6.$$

Backward: the subtraction operation

Subtraction operation $e = a - y$, locals $\partial e / \partial a = 1$, $\partial e / \partial y = -1$:

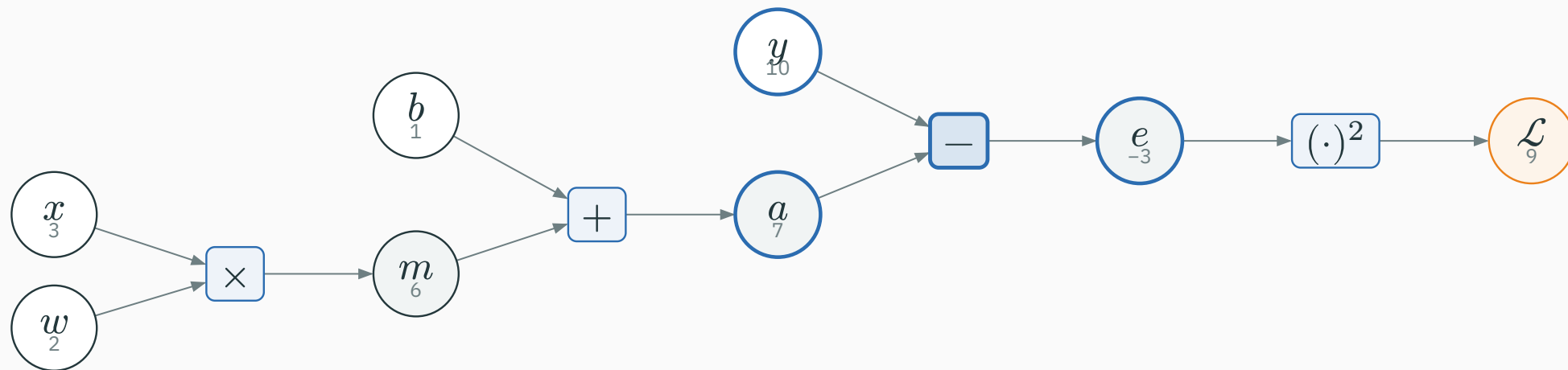
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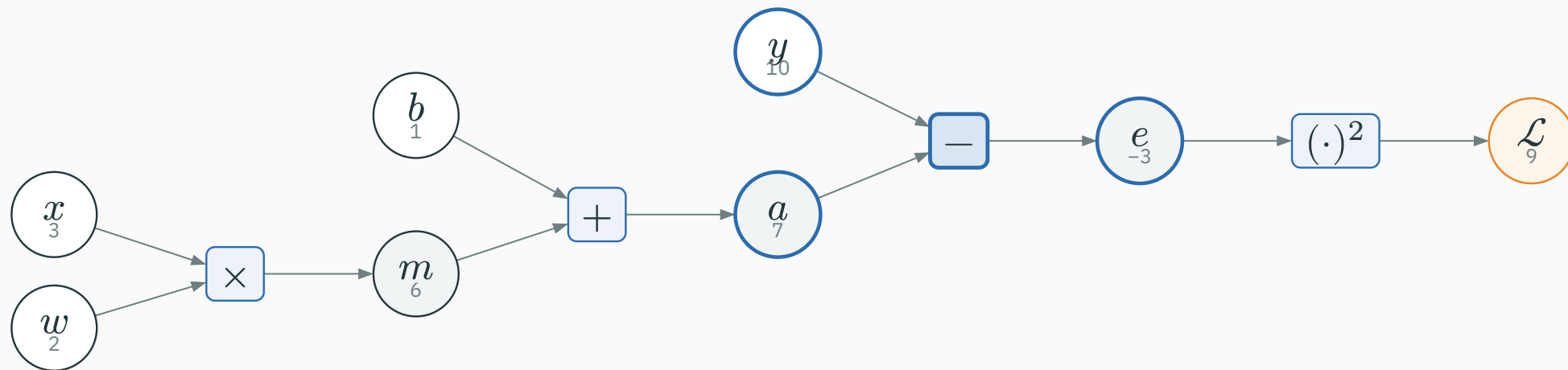
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$$\frac{\partial \mathcal{L}}{\partial a} = \frac{\partial \mathcal{L}}{\partial e} \cdot 1 = -6$$

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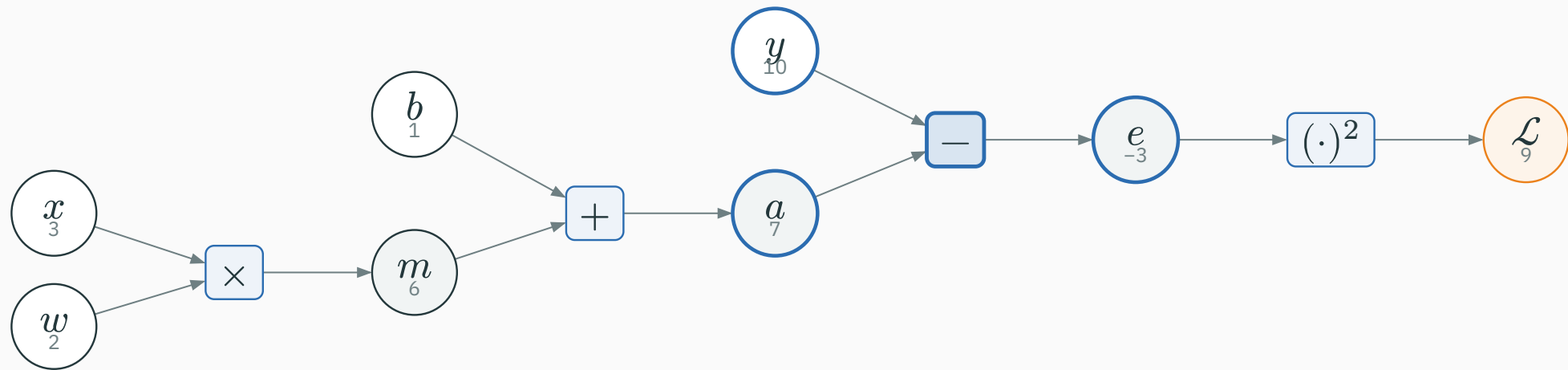
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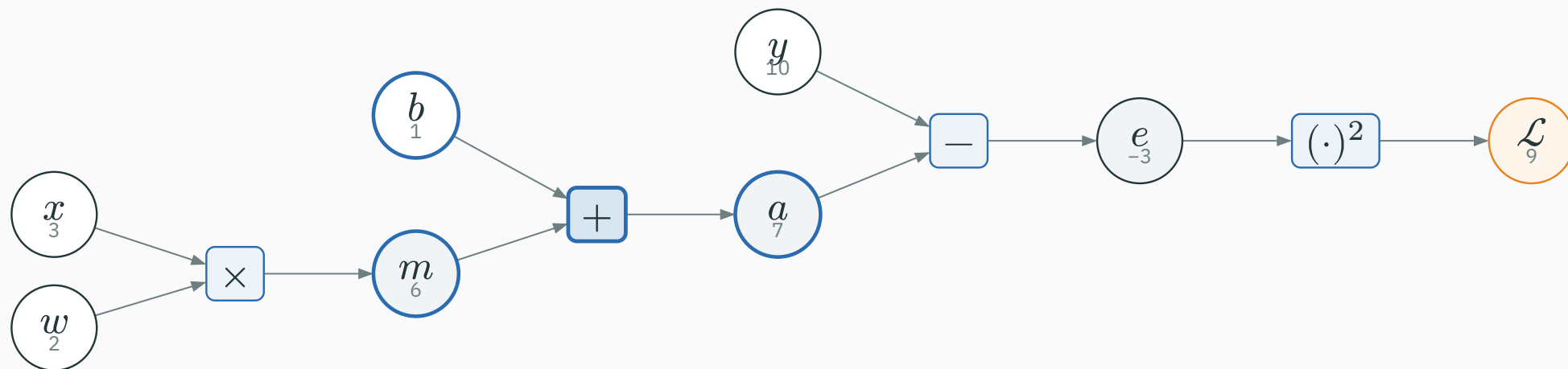
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The target y also receives a derivative. We would use $\partial \mathcal{L} / \partial y$ if y were itself a learned quantity.

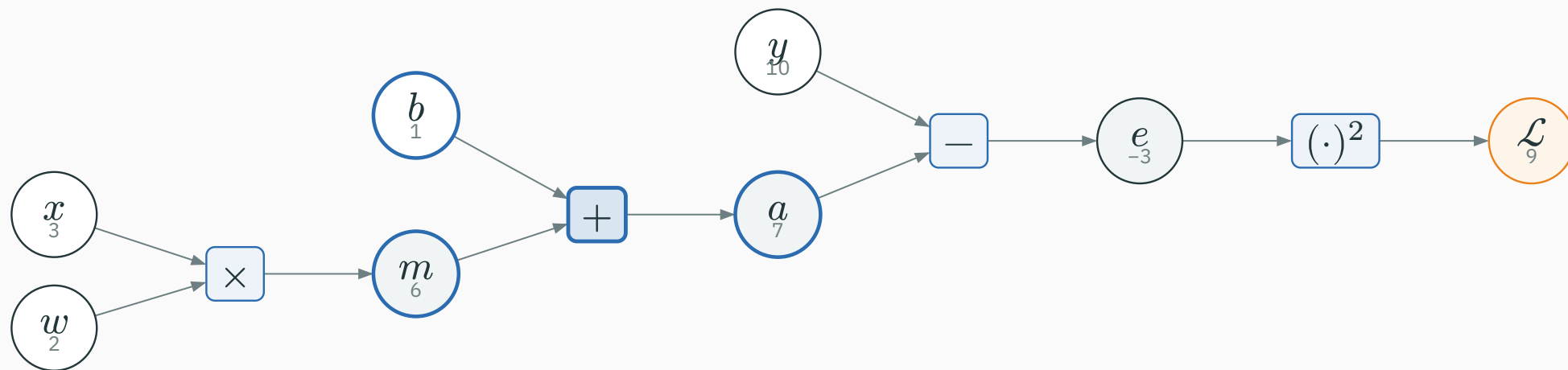
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Backward: the addition operation

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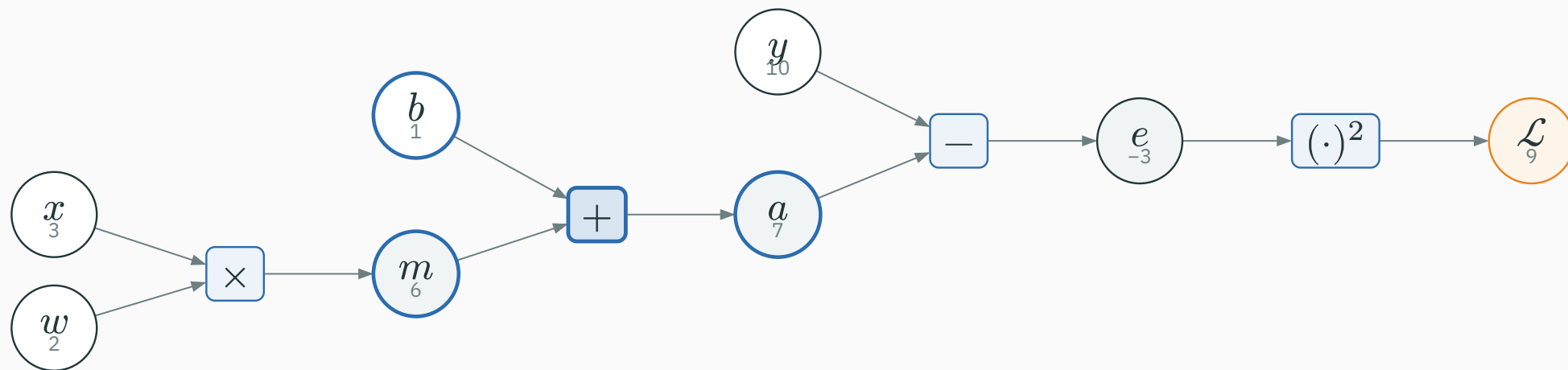


Addition operation $a = m + b$ **copies** its gradient:



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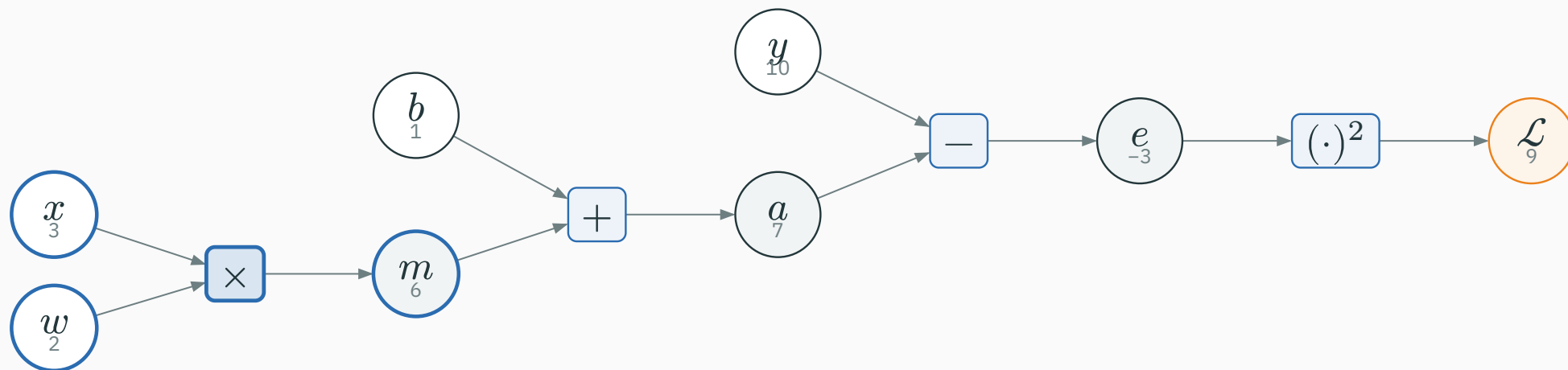
So $\partial \mathcal{L} / \partial b = -6$ is one of our final answers.

Backward: the multiplication operation

Multiplication operation $m = wx$ **sends the other input:**

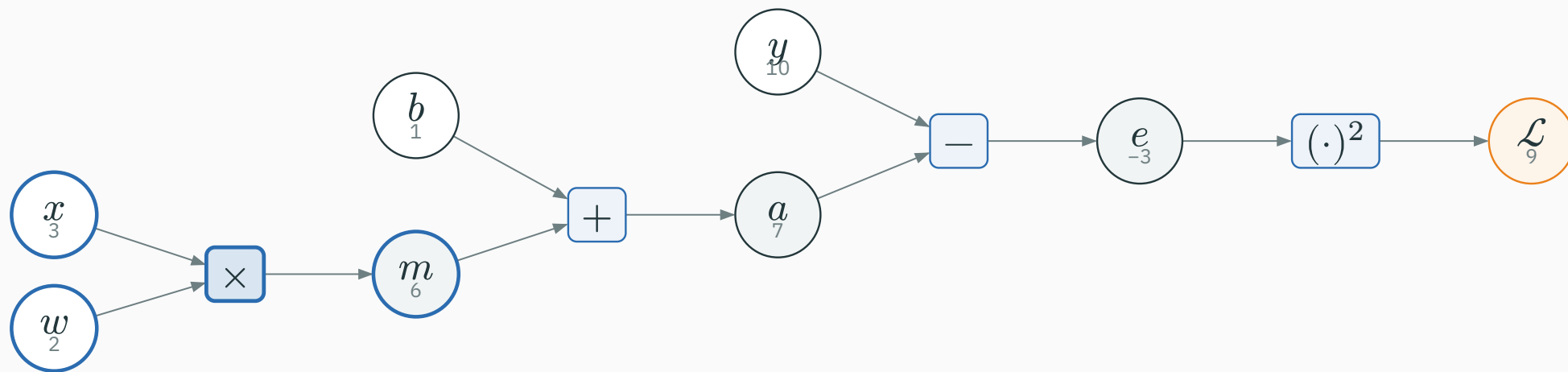
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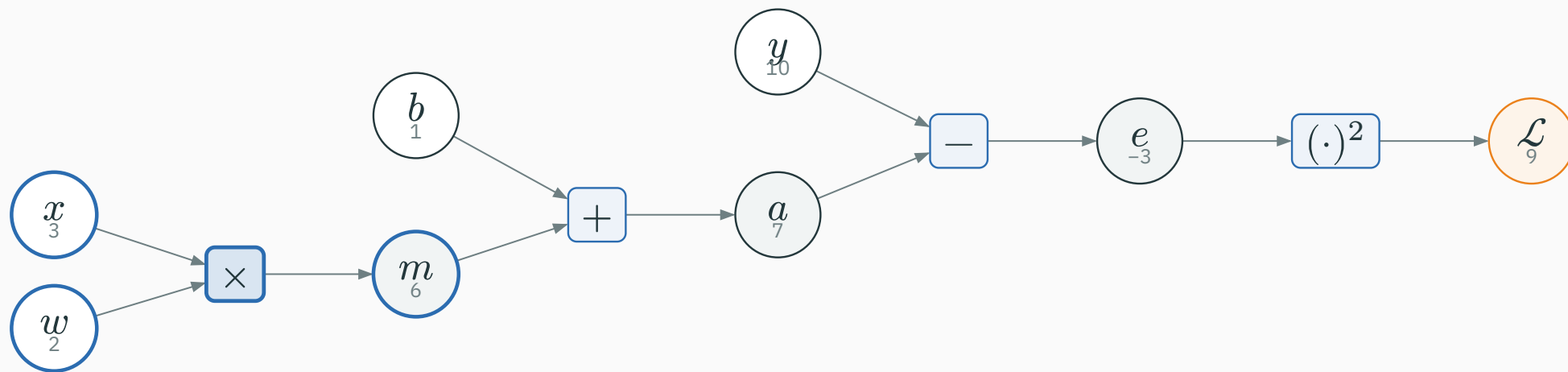
Multiplication operation $m = wx$ **sends the other input:**



$$\frac{\partial \mathcal{L}}{\partial w} = \frac{\partial \mathcal{L}}{\partial m} \cdot x = -6 \cdot 3 = -18$$

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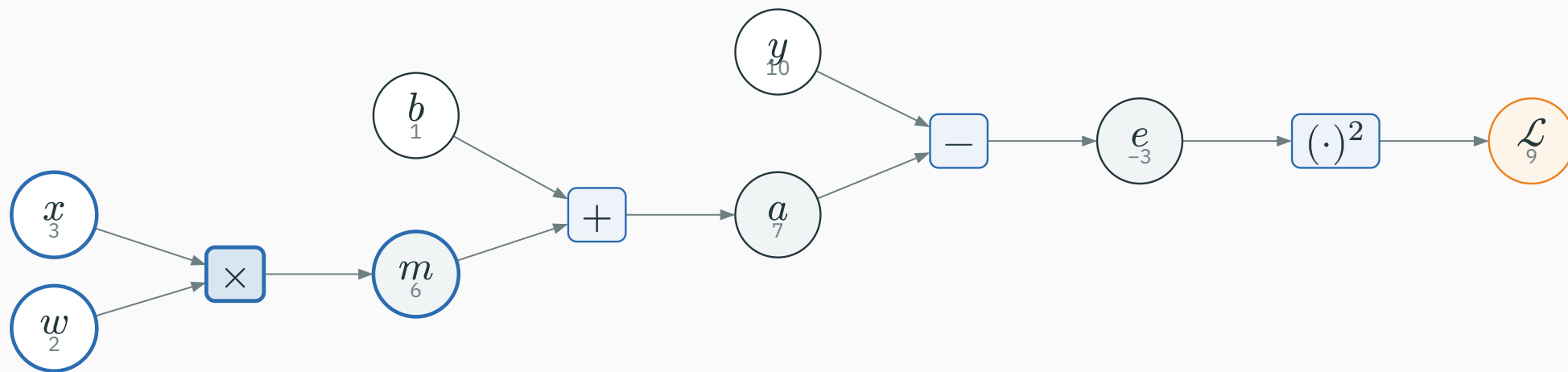


$$\frac{\partial \mathcal{L}}{\partial w} = \frac{\partial \mathcal{L}}{\partial m} \cdot x = -6 \cdot 3 = -18$$

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{\partial \mathcal{L}}{\partial m} \cdot w = -6 \cdot 2 = -12$$

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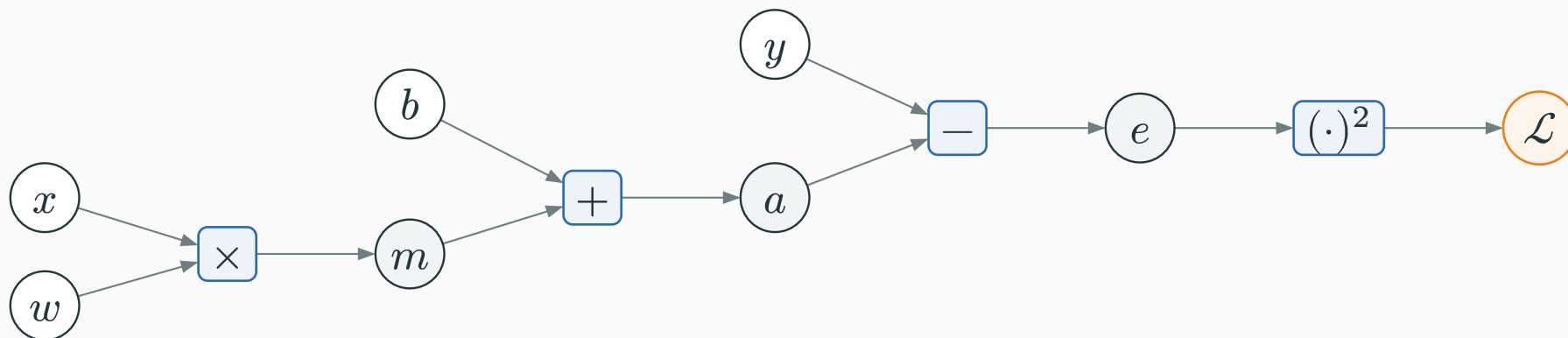
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$\partial \mathcal{L} / \partial w$	$\partial \mathcal{L} / \partial b$	$\partial \mathcal{L} / \partial x$	$\partial \mathcal{L} / \partial y$
-18	-6	-12	6

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One backward sweep produced **all four** partial derivatives — no formula re-derivation per parameter.

All stored values and their gradients



first half	w	x	m	b
forward value	2	3	6	1
gradient	$g_w = -18$	$g_x = -12$	$g_m = -6$	$g_b = -6$

second half	a	y	e	$\mathcal{L}$
forward value	7	10	-3	9
gradient	$g_a = -6$	$g_y = 6$	$g_e = -6$	$g_{\mathcal{L}} = 1$

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Backprop and the symbolic formulas agree at this point.

```
import torch

w = torch.tensor(2.0, requires_grad=True)
x = torch.tensor(3.0, requires_grad=True)
b = torch.tensor(1.0, requires_grad=True)
# Targets normally do not track gradients; enable it here to inspect dL/dy.
y = torch.tensor(10.0, requires_grad=True)

m = w * x
a = m + b
e = a - y
L = e**2
for value in (m, a, e, L):
    value.retain_grad()      # keep intermediate grads for inspection

L.backward()
print([t.item() for t in (m, a, e, L)])
# [6.0, 7.0, -3.0, 9.0]
print([t.grad.item() for t in (L, e, a, m, w, x, b, y)])
# [1.0, -6.0, -6.0, -6.0, -18.0, -12.0, -6.0, 6.0]
```

In our graph	In PyTorch	Meaning
stored scalar value	Tensor	one stored forward value
local backward rule	grad_fn	operation that produced a non-leaf tensor
saved forward value	saved by the operation	operand needed by a local derivative
reverse sweep	L.backward()	seed at $\mathcal{L}$; visit operations in reverse order
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reverse mode = direction · backprop = algorithm · autograd = engine: record → save → replay

We found $\partial \mathcal{L} / \partial w = -18$. For a small positive learning rate, which way does gradient descent move w ?

A. Decrease w

B. Increase w

C. Leave w unchanged

D. The sign tells us nothing

Answer: a negative gradient means increase the parameter

A ANSWER

Correct: B — Increase w

Why: Gradient descent subtracts the gradient: $w \leftarrow w - \eta(-18) = w + 18\eta$.

A small gradient step reduces this loss

Take $\eta = 0.01$ and update:

$$w \leftarrow 2 - 0.01(-18) = 2.18, \quad b \leftarrow 1 - 0.01(-6) = 1.06.$$

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loss falls from 9 to $(7.60 - 10)^2 = 5.76$

Backprop supplied the direction. The learning rate controls how far we move; a later optimization lecture studies that choice.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/autograd

Build a scalar computation graph, run the forward pass, then click **backward** to watch each $\partial\mathcal{L}/\partial u$ fill in — exactly the $(wx + b - y)^2$ example above.

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Trace which local rule changes when the final squared-error operation is replaced with $|e|$.

See it move

Step through values and loss derivatives in the interactive graph ↗.

PyTorch → build autograd

Open the scalar-engine Colab ↓: use plain PyTorch, trace every edge, then compare fused and atomic sigmoid.

Continue through the lecture

Use the complete Colab ↓ for branches, vectors, the dense VJP, batches, one update, and the tiny MLP.

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Easy	change one input and recompute the forward pass
Medium	derive all four gradients before calling <code>backward()</code>
Hard	replace the square by $ e $; state what happens at $e = 0$

From scalars to vectors

We store gradients as column vectors. The name of the local object depends on the input and output shapes:

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Operation shape	Local object	Name
$u \in \mathbb{R} \rightarrow v \in \mathbb{R}$	$\partial v / \partial u \in \mathbb{R}$	derivative
$\mathbf{x} \in \mathbb{R}^d \rightarrow z \in \mathbb{R}$	$\partial z / \partial \mathbf{x} \in \mathbb{R}^d$	gradient
$\mathbf{x} \in \mathbb{R}^d \rightarrow \mathbf{z} \in \mathbb{R}^m$	$J_{ij} = \partial z_i / \partial x_j$	Jacobian $\mathbf{J} \in \mathbb{R}^{m \times d}$

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a Jacobian contains one local derivative for every output–input pair

A 2×2 Jacobian is four familiar derivatives

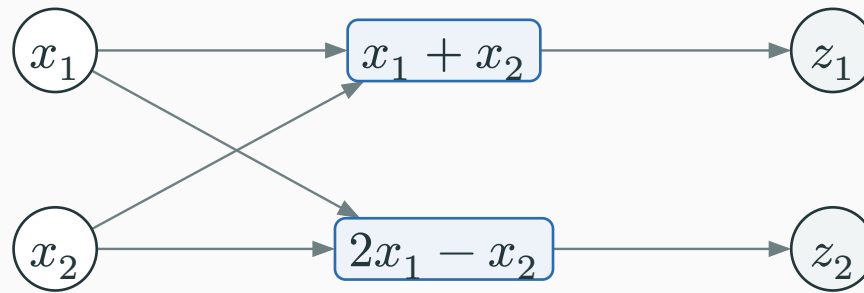
Take a vector operation with two inputs and two outputs:

$$z_1 = x_1 + x_2, \quad z_2 = 2x_1 - x_2.$$

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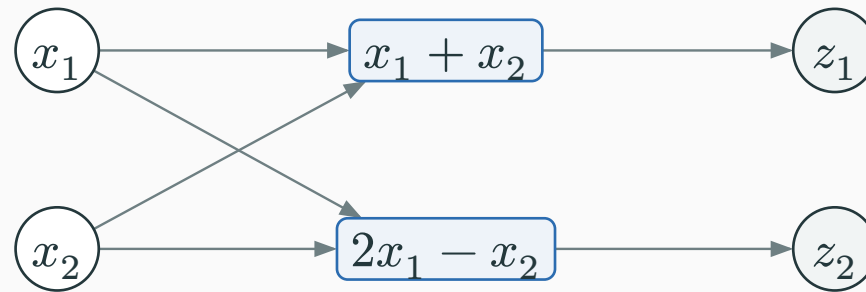
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$$z_1 = x_1 + x_2, \quad z_2 = 2x_1 - x_2.$$

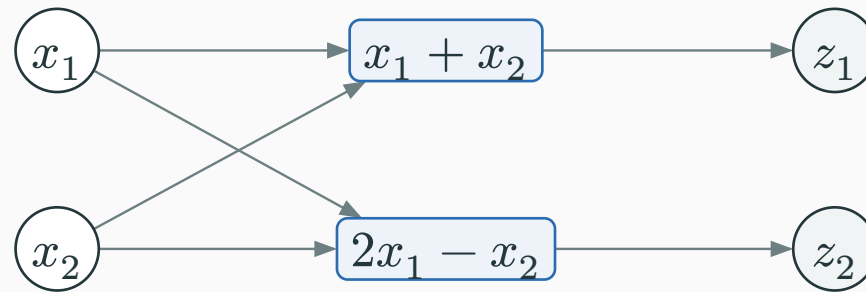


Output	Input x_1	Input x_2
z_1	$\partial z_1 / \partial x_1 = 1$	$\partial z_1 / \partial x_2 = 1$
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$$\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} \cdot \text{rows are outputs} \cdot \text{columns are inputs}$$

Vector chain rule: make the shapes fit

For $x \in \mathbb{R}^d \rightarrow z \in \mathbb{R}^m$, let the **local Jacobian** be

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| When $d = m = 1$, J is one number and this is the scalar chain rule from earlier.

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the vector rule is the scalar branch rule applied to every input

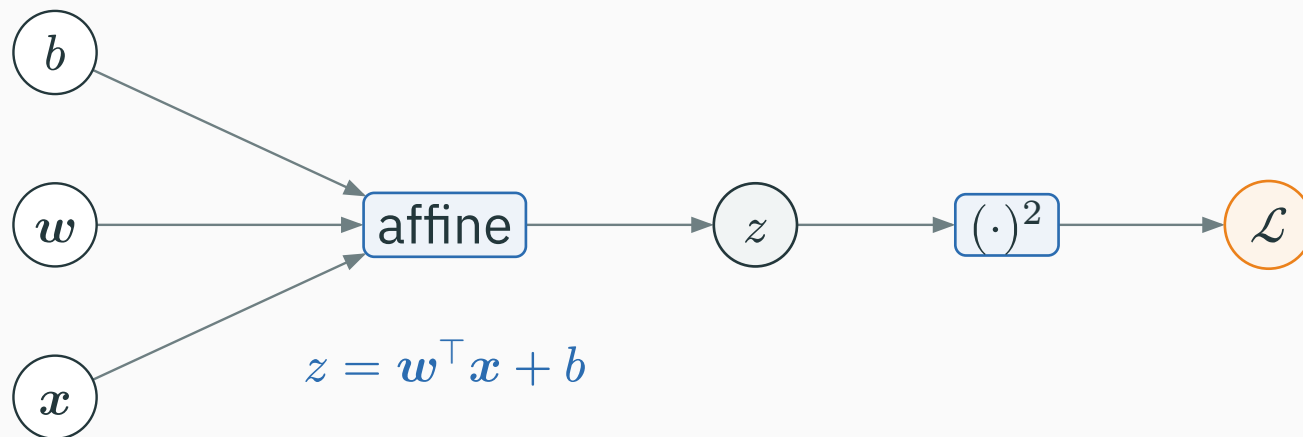
Affine means a weighted sum plus a bias:

$$z = \mathbf{w}^\top \mathbf{x} + b = \sum_{j=1}^d w_j x_j + b.$$

An affine operation adds scalar products and a bias

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$x_i \rightarrow x_i + \Delta$	$w_i(x_i + \Delta) - w_i x_i = w_i \Delta$	$\partial z / \partial x_i = w_i$
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local derivative = change in the operation's output $\div$ change in one input

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the vector formulas are just the scalar coordinate derivatives stacked together

Affine forward: multiply componentwise, then add

Use $x = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$, $w = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$, and $b = 0$.

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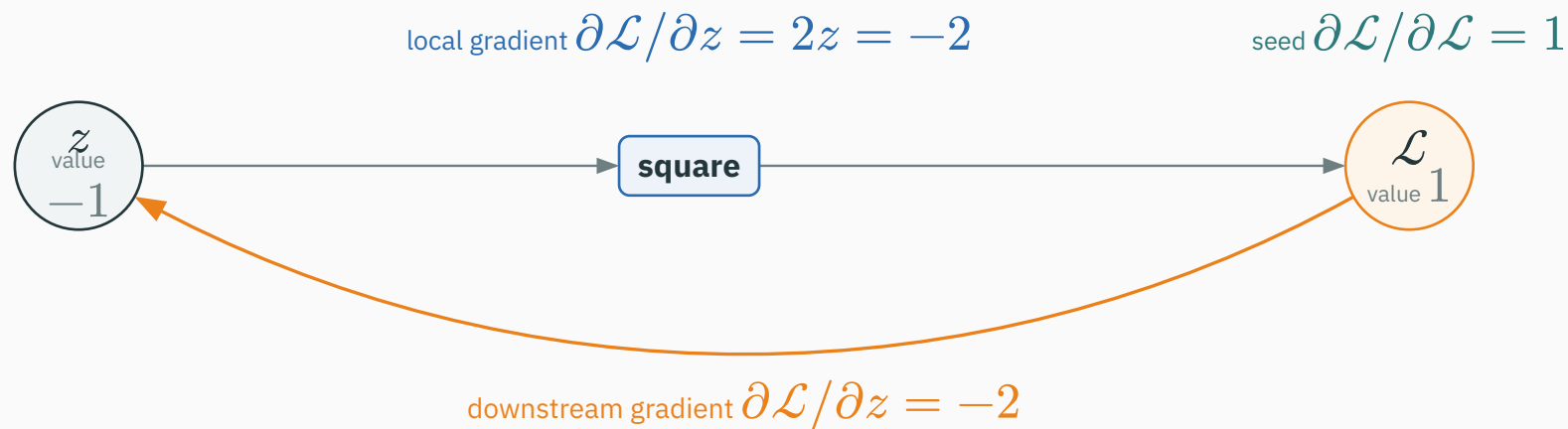
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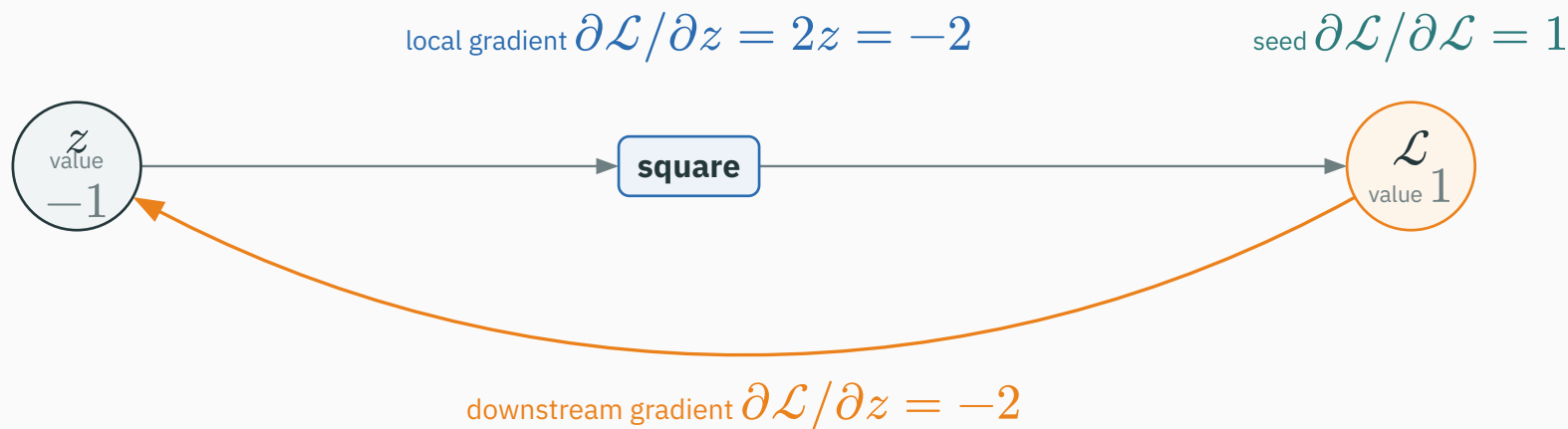
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dot product = multiply matching entries, then sum

First backward step: the square operation



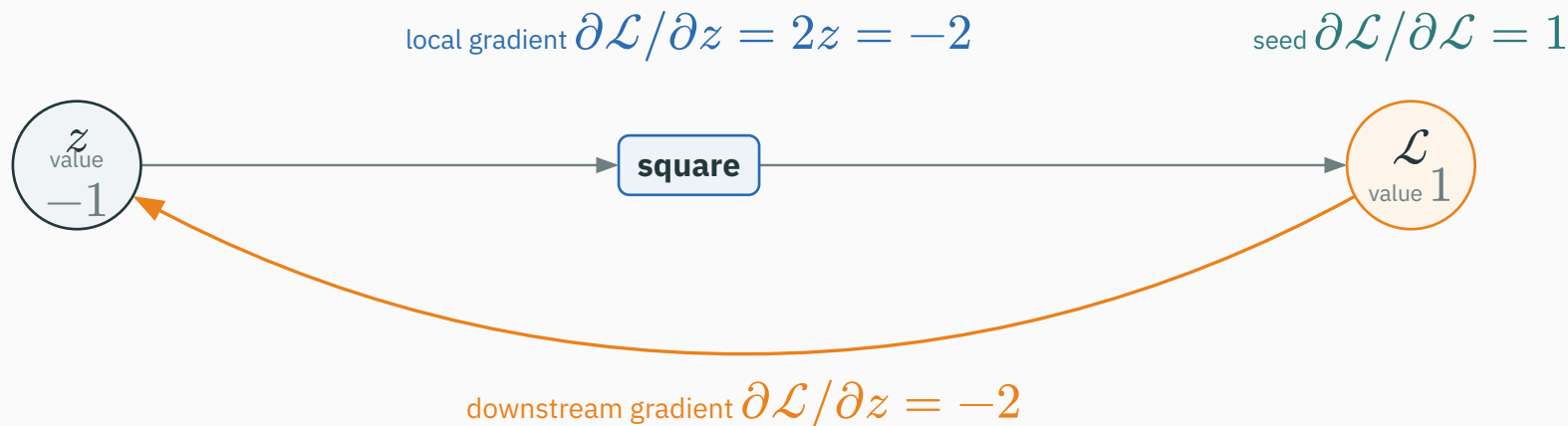
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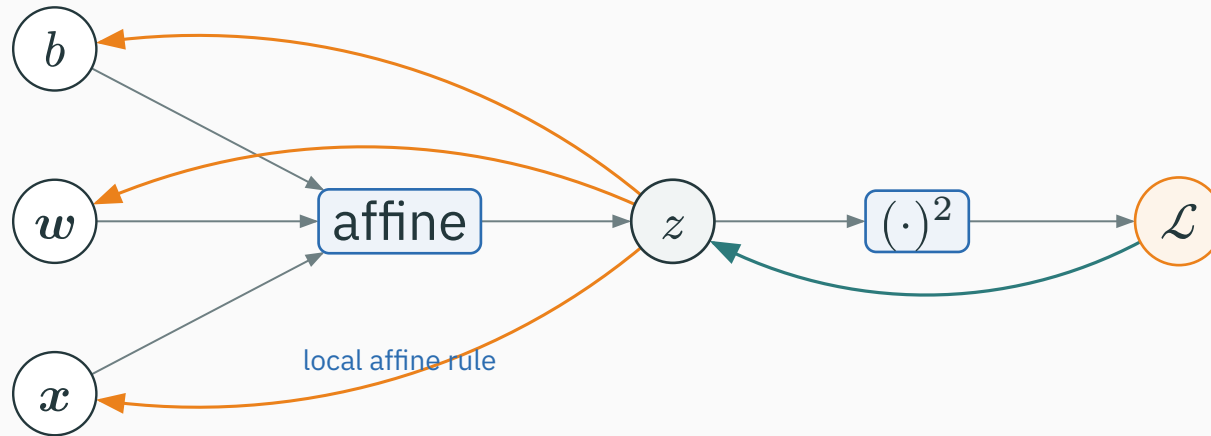


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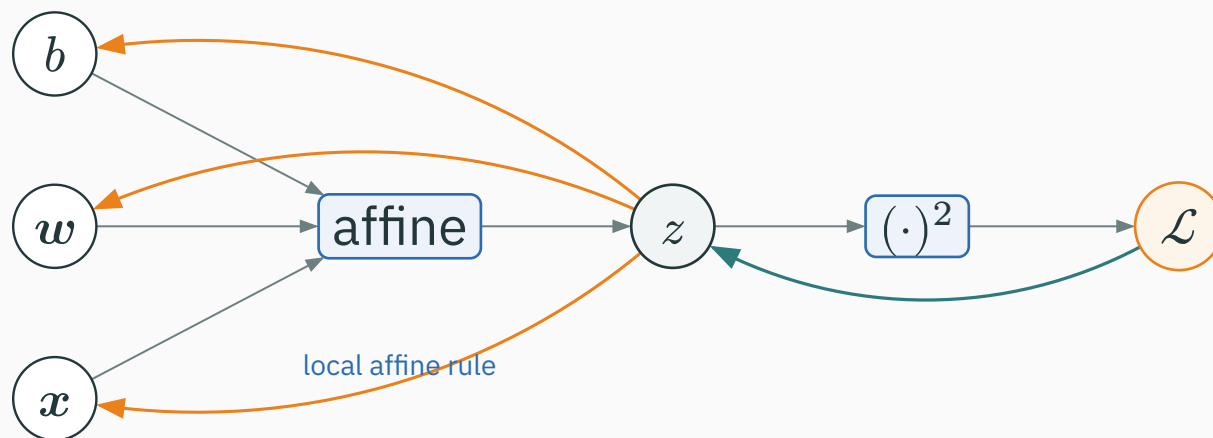
$$\frac{\partial \mathcal{L}}{\partial z} = 1 \cdot 2z = -2.$$

the square returns -2 ; it now arrives at the affine operation

Affine backward: apply the three local gradients

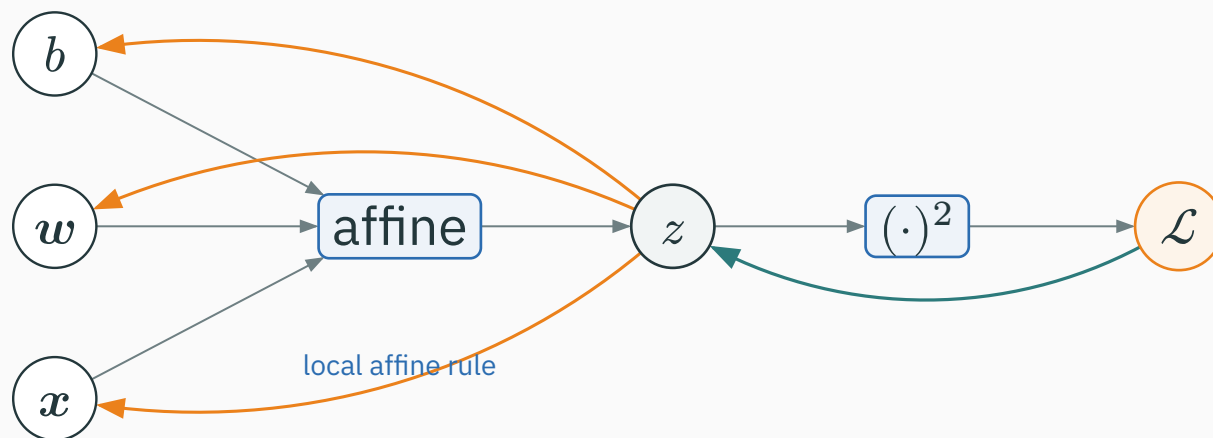


Affine backward: apply the three local gradients



Let $g_z = \partial \mathcal{L} / \partial z = -2$ arrive at the affine operation. Its **local gradients** send back:

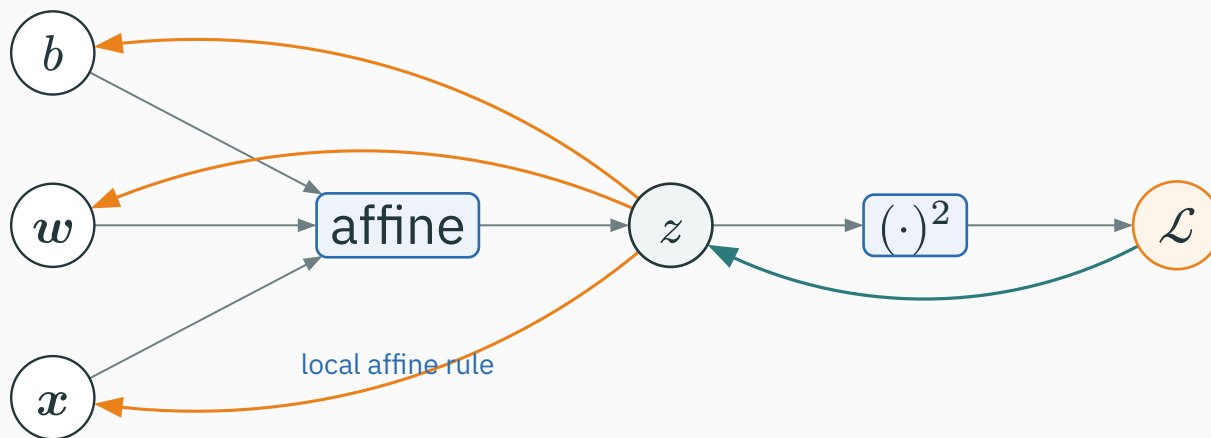
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Input	Local gradient	Downstream gradient
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x	$\partial z / \partial x = w$	$\partial \mathcal{L} / \partial x = (-2)w = (-2, -6)^\top$
b	$\partial z / \partial b = 1$	$\partial \mathcal{L} / \partial b = -2$

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one arriving gradient g_z produces three separate downstream gradients

PyTorch gives the same vector gradients

```
import torch

x = torch.tensor([2., -1.], requires_grad=True)
w = torch.tensor([1., 3.], requires_grad=True)
b = torch.tensor(0., requires_grad=True)

z = w @ x + b
L = z**2
L.backward()

print(z.item(), L.item())  # -1.0, 1.0
print(w.grad)              # tensor([-4.,  2.])
print(x.grad, b.grad)     # tensor([-2., -6.]), tensor(-2.)
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@ performs the dot product; backward() still applies the same local rules

For this one-output affine node, PyTorch sends the **upstream gradient** `grad_z` into a backward rule. A readable teaching equivalent is:

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class Affine(torch.autograd.Function):
    @staticmethod
    def forward(ctx, x, w, b):
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        return w @ x + b

    @staticmethod
    def backward(ctx, grad_z):        # grad_z = dL/dz
        x, w = ctx.saved_tensors
        grad_x = grad_z * w          # dL/dx = dL/dz * dz/dx
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        grad_b = grad_z              # dL/db = dL/dz * 1
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Teaching equivalent. `nn.Linear.forward` calls `F.linear`; production autograd uses generated native kernels. Linear source ↗ · derivative rules ↗

From one scalar output to a vector output

One neuron still produces one scalar

Start with the affine rule we already know:

$$z = \boldsymbol{w}^\top \boldsymbol{x} + b.$$

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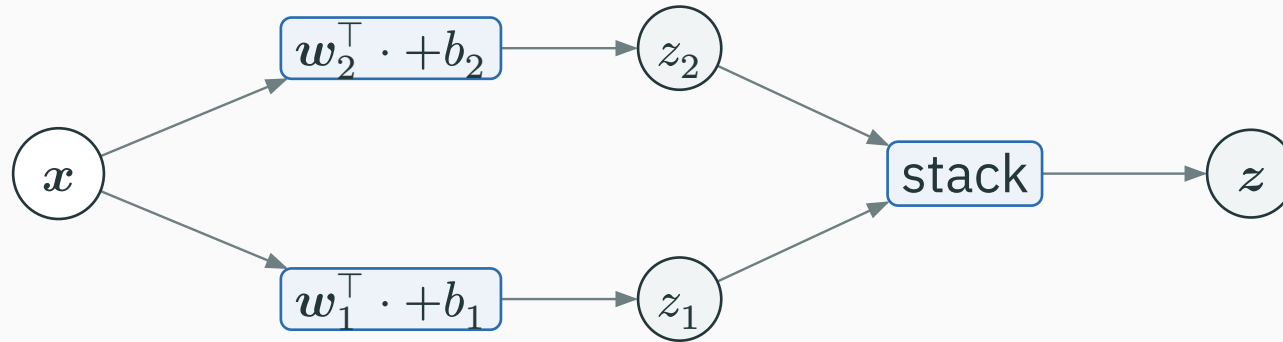
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one weight row dotted with one input vector produces one number

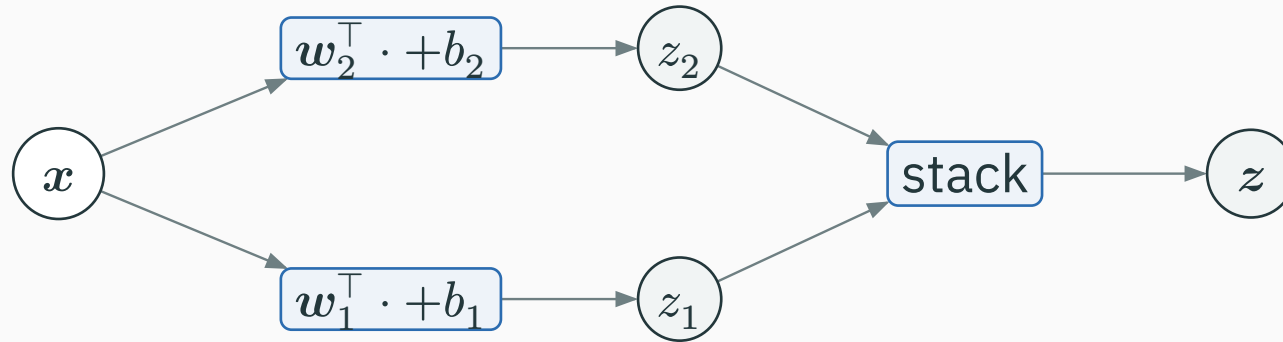
Two neurons produce two scalars

Give each neuron its own weight row and bias; both read the same x .



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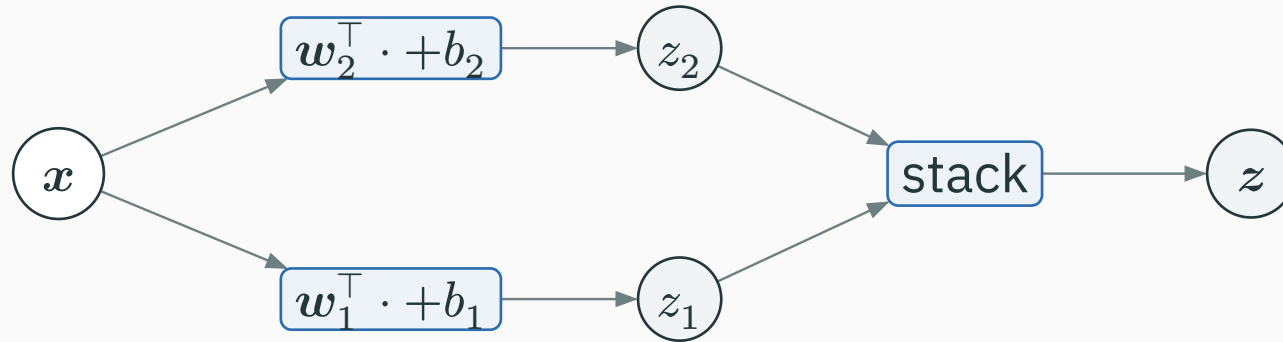
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$$z_1 = w_1^\top x + b_1, \quad z_2 = w_2^\top x + b_2.$$

stack the two numbers into $z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}$

Stacking the two neurons gives one dense layer

D DERIVATION

$$\mathbf{W} = \begin{pmatrix} \mathbf{w}_1^\top \\ \mathbf{w}_2^\top \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}, \quad z = \mathbf{W}x + b.$$

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| d is the number of input features; m is the number of output neurons.

Use

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multiply matching entries, add them, then add the bias

Work out output 2, then stack

The second neuron uses a different row:

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$$\mathbf{z} = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} -1 \\ -4 \end{pmatrix}, \quad \mathbf{W} = \begin{pmatrix} 1 & 3 \\ -2 & 1 \end{pmatrix}.$$

The second neuron uses a different row:

$$\mathbf{w}_2^\top = (-2, 1), \quad b_2 = 1.$$

$$z_2 = (-2)(2) + 1(-1) + 1 = -4.$$

$$\mathbf{z} = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} -1 \\ -4 \end{pmatrix}, \quad \mathbf{W} = \begin{pmatrix} 1 & 3 \\ -2 & 1 \end{pmatrix}.$$

matrix notation only stacks the two familiar dot products

Backward starts with one arrival per output

Suppose the later loss returns

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Read this vector as two scalar messages:

Output	Arriving gradient	Meaning
z_1	$g_1 = \partial \mathcal{L} / \partial z_1 = 4$	nudge z_1 upward; loss rises at rate 4
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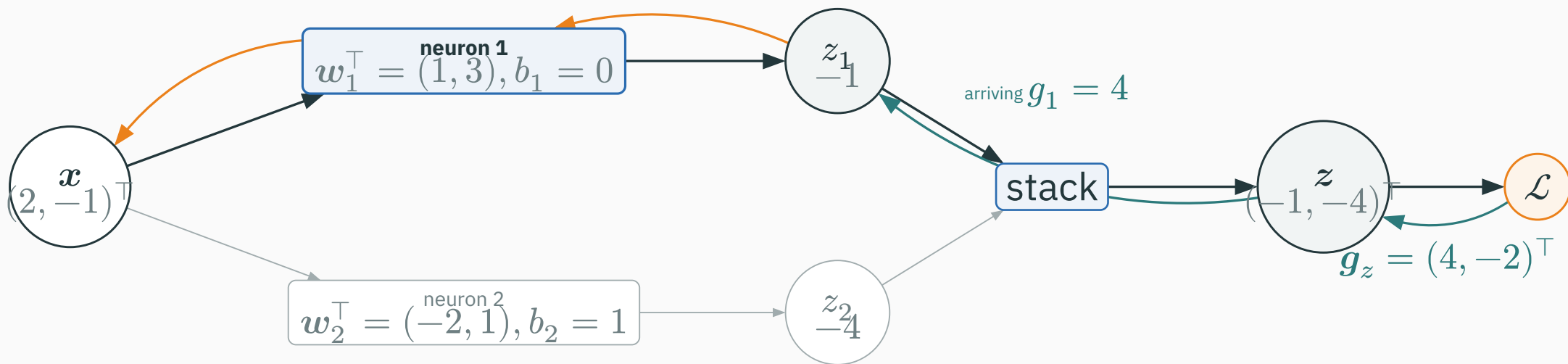
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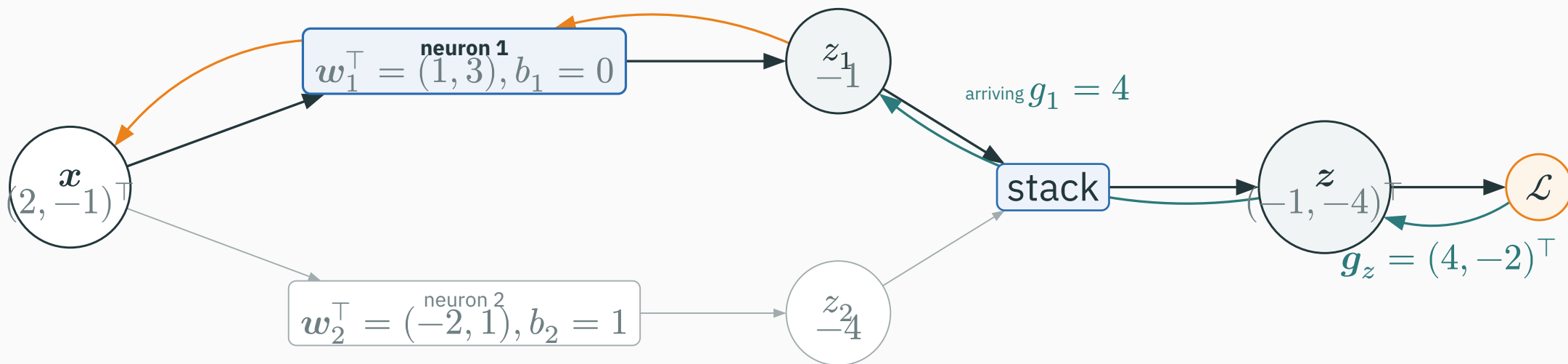
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a vector gradient is simply one arriving number for each output coordinate

Backward through neuron 1: trace one scalar path

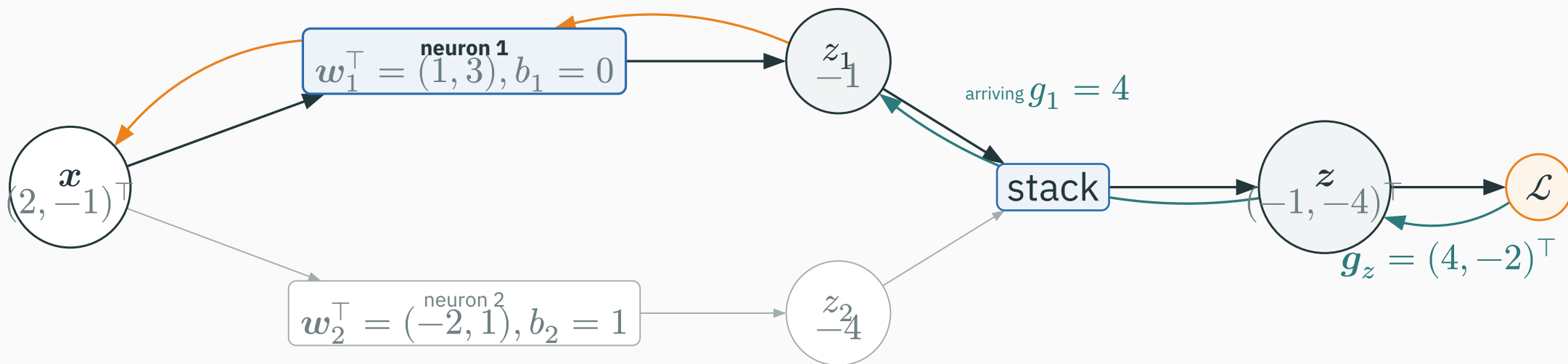


Backward through neuron 1: trace one scalar path



$$\mathbf{W}_{[1,:]} = \mathbf{w}_1^\top = (1, 3) \quad \mathbf{b}_{[1]} = b_1 = 0 \quad \mathbf{z}_{[1]} = z_1 = -1 \quad \mathbf{g}_{z[1]} = g_1 = 4$$

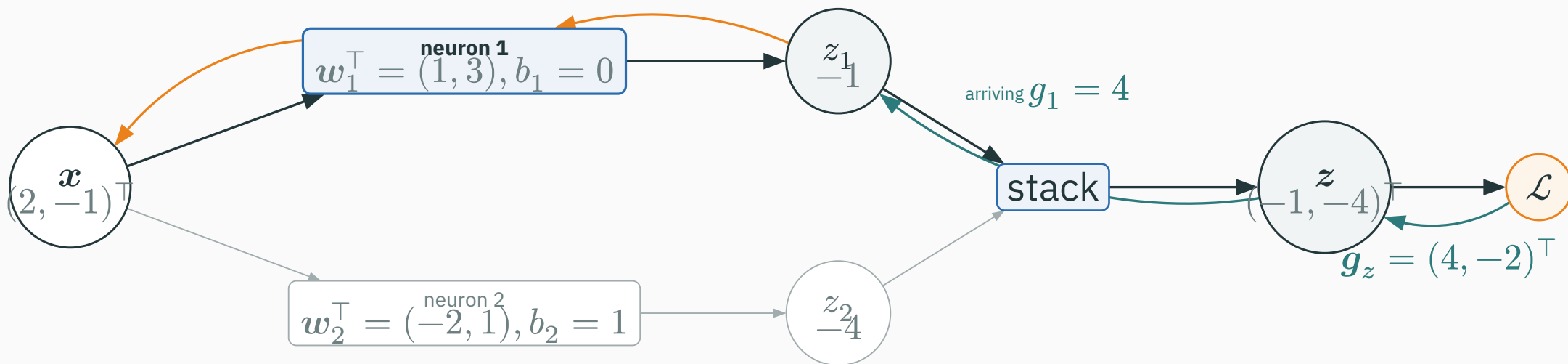
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one matrix row and one vector coordinate recover the scalar neuron we already know

Neuron 1 returns gradients to its row, bias, and input

D DERIVATION

The highlighted node receives $g_1 = 4$. Reuse the scalar affine rule:

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Recipient	Local derivative	Gradient returned
w_1	$\partial z_1 / \partial w_1 = x$	$g_1 x = 4 \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 8 \\ -4 \end{pmatrix}$
b_1	$\partial z_1 / \partial b_1 = 1$	$g_1 = 4$
x	$\partial z_1 / \partial x = w_1$	$g_1 w_1 = 4 \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 12 \end{pmatrix}$

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neuron 1 updates only row 1 of W , but it also returns a contribution to the shared input

Neuron 2 receives $g_2 = -2$. Apply the same rule again:

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Recipient	Local derivative	Gradient returned
w_2	$\partial z_2 / \partial w_2 = x$	$g_2 x = (-2)(2, -1)^\top = (-4, 2)^\top$
b_2	$\partial z_2 / \partial b_2 = 1$	$g_2 = -2$
x	$\partial z_2 / \partial x = w_2$	$g_2 w_2 = (-2)(-2, 1)^\top = (4, -2)^\top$

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x	$\partial z_2 / \partial x = w_2$	$g_2 w_2 = (-2)(-2, 1)^\top = (4, -2)^\top$

neuron 2 updates row 2 and returns its own contribution to x

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The transpose lines up the two output contributions for each input coordinate; the matrix multiplication performs the required addition.

Parameter gradients keep the two rows separate

D DERIVATION

$$\mathbf{g}_W = \begin{pmatrix} 8 & -4 \\ -4 & 2 \end{pmatrix}, \quad \mathbf{g}_b = \begin{pmatrix} 4 \\ -2 \end{pmatrix}.$$

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one output coordinate creates one row of the weight gradient

Dense backward: three rules, with shapes

Returned to	Rule	Shape
x	$g_x = W^\top g_z$	$(d \times 1) = (d \times m)(m \times 1)$
W	$g_W = g_z x^\top$	$(m \times d) = (m \times 1)(1 \times d)$
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Do the scalar reasoning first. Use the shapes as a check that the stacked formula has the correct orientation.

PyTorch stores gradients in matching shapes

```
import torch

x = torch.tensor([2., -1.], requires_grad=True)           # (2,)
W = torch.tensor([[1., 3.], [-2., 1.]],                  # (2, 2)
                  requires_grad=True)
b = torch.tensor([0., 1.], requires_grad=True)            # (2,)

z = W @ x + b                                              # (2,)
z.backward(torch.tensor([4., -2.]))                       # arriving g_z

print(x.grad)  # tensor([ 8., 10.])
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```

every tensor and its gradient have the same shape

From one example to a batch

A batch is a stack of examples

Use two examples so every calculation remains visible:

$$\mathbf{X} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

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$$\mathbf{X} \in \mathbb{R}^{B \times d} = \mathbb{R}^{2 \times 2}$$

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the new first axis counts examples; it does not create new parameters

The same dense layer processes both rows

Keep the same parameters

$$\mathbf{W} = \begin{pmatrix} 1 & 3 \\ -2 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

The same dense layer processes both rows

D DERIVATION

Keep the same parameters

$$\mathbf{W} = \begin{pmatrix} 1 & 3 \\ -2 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$
$$\mathbf{Z} = \mathbf{X}\mathbf{W}^\top + \mathbf{1}_B\mathbf{b}^\top = \begin{pmatrix} -1 & -4 \\ 5 & 5 \end{pmatrix}.$$

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each row runs the same forward calculation with the same $\mathbf{W}$, $\mathbf{b}$

Two outputs mean a length-2 target per example

Let $n \in \{1, 2\}$ be the **example (row) index**; the parenthesized superscript (n) is a label, not a power. Here $m = 2$, so $\mathbf{y}^n \in \mathbb{R}^2$.

$$\mathbf{Y} = \begin{pmatrix} (\mathbf{y}^1)^\top \\ (\mathbf{y}^2)^\top \end{pmatrix} = \begin{pmatrix} -5 & -2 \\ 7 & 1 \end{pmatrix} \in \mathbb{R}^{B \times m} = \mathbb{R}^{2 \times 2}.$$

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Example index n	$\mathbf{z}^n$	$\mathbf{y}^n$	$\mathbf{r}^n = \mathbf{z}^n - \mathbf{y}^n$
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each $\mathbf{r}^n$ is a vector; squaring and summing its coordinates gives scalar ℓ_n

Scalar-output special case: if $m = 1$, each target $\mathbf{y}^n$ is a scalar and $\mathbf{Y}$ has shape $B \times 1$.

The mean scales each example's residual by $\frac{1}{2}$

D DERIVATION

First recompute the two residual vectors from the previous slide:

$$\mathbf{r}^1 = (-1, -4) - (-5, -2) = (4, -2)$$

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Stack the residuals as rows:

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| Row n is $\frac{1}{2}\mathbf{r}^n$. Rows are examples; columns are output neurons.

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Before averaging, example 1 has $r^1 = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$ and $x^1 = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$.

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this is the same single-example outer product derived on the previous slides

Example 2 proposes another gradient

Example 2 has $r^2 = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$ and $x^2 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$.

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Both examples used the same parameters, so their paths add; the mean divides by 2:

$$\mathbf{g}_W = \frac{1}{2}(\mathbf{G}_W^1 + \mathbf{G}_W^2) = \begin{pmatrix} 5 & -4 \\ -4 & 5 \end{pmatrix}.$$

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$$\mathbf{g}_b = \frac{1}{2} \left(\begin{pmatrix} 4 \\ -2 \end{pmatrix} + \begin{pmatrix} -2 \\ 4 \end{pmatrix} \right) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

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$$\mathbf{g}_W = \mathbf{G}_Z^\top \mathbf{X}.$$

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$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 5 & -4 \\ -4 & 5 \end{pmatrix}.$$

The middle dimension B disappears because matrix multiplication sums the contribution from every example.

Batch gradients have one simple shape story

Returned to	Rule	Shape
W	$g_W = G_Z^\top X$	$(m \times d) = (m \times B)(B \times d)$
b	$g_b = G_Z^\top \mathbf{1}_B$	$(m \times 1) = (m \times B)(B \times 1)$
X	$g_X = G_Z W$	$(B \times d) = (B \times m)(m \times d)$

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$$g_X = \begin{pmatrix} 4 & 5 \\ -5 & -1 \end{pmatrix}.$$

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$$g_X = \begin{pmatrix} 4 & 5 \\ -5 & -1 \end{pmatrix}.$$

shared parameter gradients add across rows; input gradients remain one row per example

Sum and mean differ only by scale

Reduction	Arriving gradient	Weight gradient
sum	$G_Z = R$	$g_W = R^\top X$
mean	$G_Z = \frac{R}{B}$	$g_W = R^\top \frac{X}{B}$

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```
loss = 0.5 * ((Z - Y)**2).sum(dim=1).mean()  
# sum output coordinates inside each example; mean over examples
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```
loss = 0.5 * ((Z - Y)**2).sum(dim=1).mean()  
# sum output coordinates inside each example; mean over examples
```

the dense-layer rule is unchanged; the final scalar reduction chooses the scale

PyTorch reproduces the two-example calculation

```
X = torch.tensor([[ 2., -1.], [-1., 2.]], requires_grad=True) # (B=2, d=2)
Y = torch.tensor([[-5., -2.], [ 7., 1.]]) # (2, 2)
W = torch.tensor([[1., 3.], [-2., 1.]], requires_grad=True) # (m=2, d=2)
b = torch.tensor([0., 1.], requires_grad=True) # (2,)

Z = X @ W.T + b # (B, m)
loss = 0.5 * ((Z - Y)**2).sum(dim=1).mean() # scalar
loss.backward()

print(loss.item()) # 10.0
print(W.grad) # [[ 5., -4.], [-4., 5.]]
print(b.grad) # [1., 1.]
print(X.grad) # [[ 4., 5.], [-5., -1.]]
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I INTERACTIVE

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print(W.grad) # [[ 5., -4.], [-4., 5.]]
print(b.grad) # [1., 1.]
print(X.grad) # [[ 4., 5.], [-5., -1.]]
```

backward() carries out the same rowwise contributions and additions

The training loop repeats this batch calculation

I INTERACTIVE

```
for X, Y in loader:
    optimizer.zero_grad()      # discard gradients from the previous update
    Z = model(X)               # batch forward: (B, d) -> (B, m)
    loss = loss_fn(Z, Y)       # reduce B examples to one scalar
    loss.backward()            # add this batch's parameter contributions
    optimizer.step()           # update parameters once
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```

clear → forward → scalar loss → backward → update

step() uses the gradients; it does not compute or clear them. That is why the next iteration begins with zero_grad().

Microbatches split one batch without changing the goal

```
optimizer.zero_grad()  
(loss_example_1 / 2).backward()    # first half of the mean gradient  
(loss_example_2 / 2).backward()    # second half accumulates  
optimizer.step()
```

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For equivalence, keep parameters fixed, preserve the same overall scaling, and do not call `zero_grad()` or `step()` between pieces.

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For equivalence, keep parameters fixed, preserve the same overall scaling, and do not call `zero_grad()` or `step()` between pieces.

one update window may contain one full batch or several correctly scaled pieces

A tiny neural network

ReLU is what prevents two dense layers from collapsing into one

Without a nonlinearity, two affine maps are still one affine map:

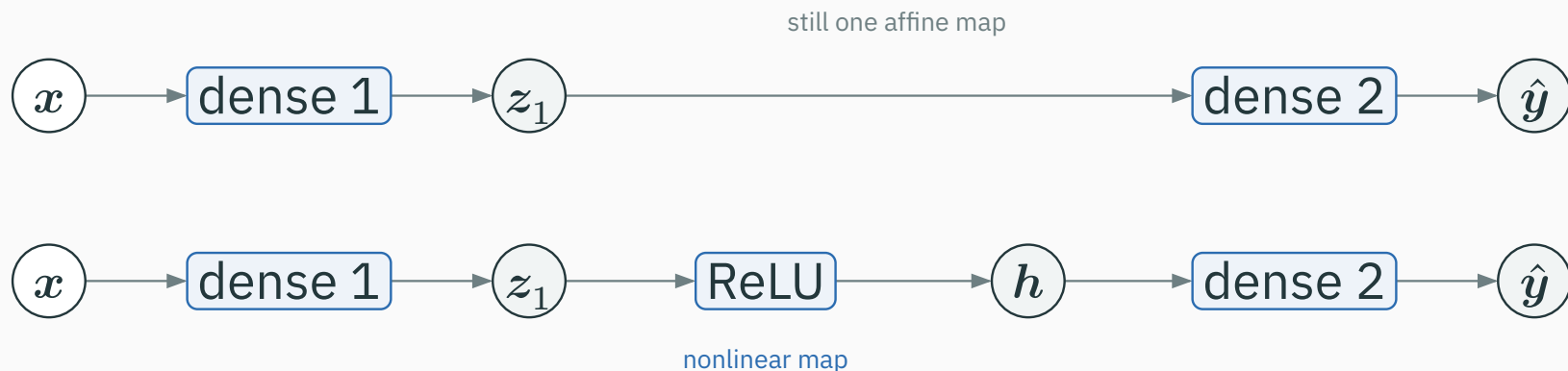
$$\begin{aligned} \mathbf{W}_2(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2 \\ = (\mathbf{W}_2\mathbf{W}_1)\mathbf{x} + (\mathbf{W}_2\mathbf{b}_1 + \mathbf{b}_2). \end{aligned}$$

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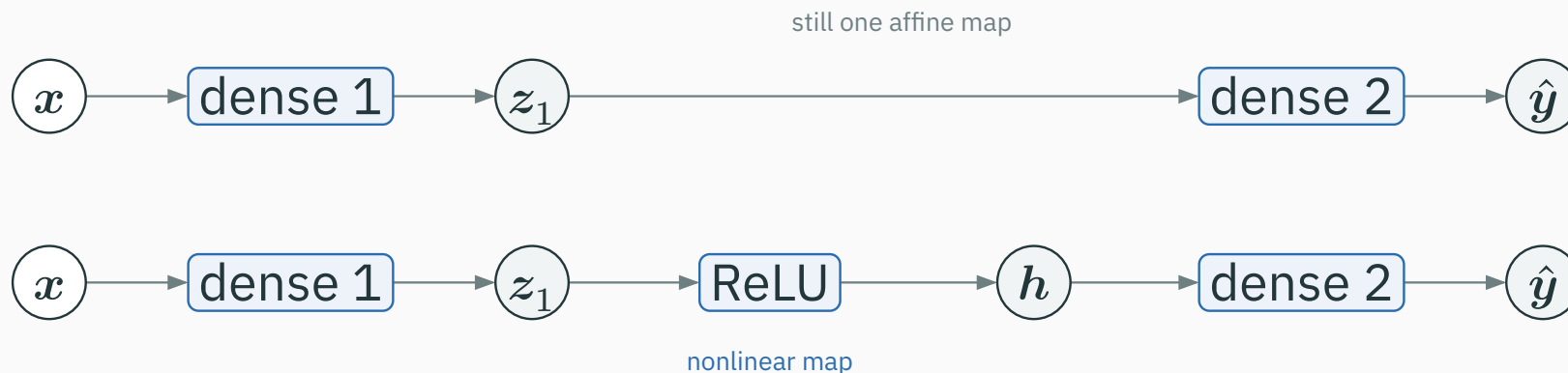


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ReLU makes the composition nonlinear; depth can now represent a richer function

A tiny MLP composes three vector blocks



A tiny MLP composes three vector blocks



Value	Rule	Shape
z_1	$W_1 x + b_1$	$(h \times d)(d \times 1) + (h \times 1) = h \times 1$
h	$\text{ReLU}(z_1)$	$h \times 1$
$\hat{y}$	$W_2 h + b_2$	$(m \times h)(h \times 1) + (m \times 1) = m \times 1$
$\mathcal{L}$	$\sum_i (\hat{y}_i - y_i)^2$	scalar

A tiny MLP composes three vector blocks



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d input features $\rightarrow h$ hidden features $\rightarrow m$ output coordinates $\rightarrow$ one loss

$$z_1 = \mathbf{W}_1 \mathbf{x} + \mathbf{b}_1 \in \mathbb{R}^{h \times 1}$$

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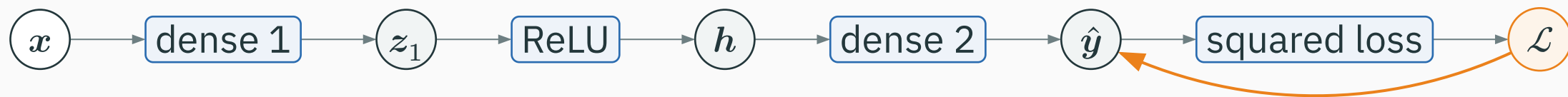
$$\hat{\mathbf{y}} = \mathbf{W}_2 \mathbf{h} + \mathbf{b}_2 \in \mathbb{R}^{m \times 1}$$

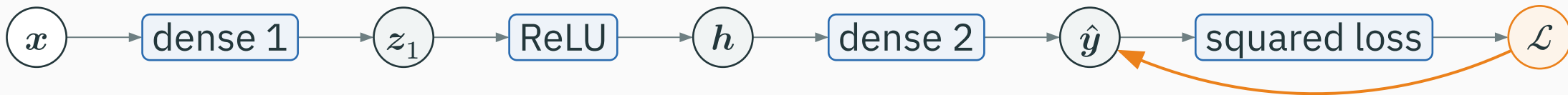
$$\mathcal{L} = \sum_i (\hat{y}_i - y_i)^2 \in \mathbb{R}$$

forward changes the representation shape; the loss finally collapses it to one number

Backward, part 1: loss $\rightarrow$ predictions

D DERIVATION

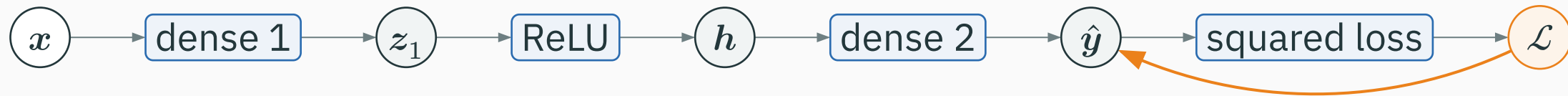




1. Squared loss. Start with the seed $\partial\mathcal{L}/\partial\mathcal{L} = 1$. The loss operation returns

$$g_{\hat{y}} = 2(\hat{y} - y).$$

$$g_{\hat{y}} \in \mathbb{R}^{m \times 1}$$

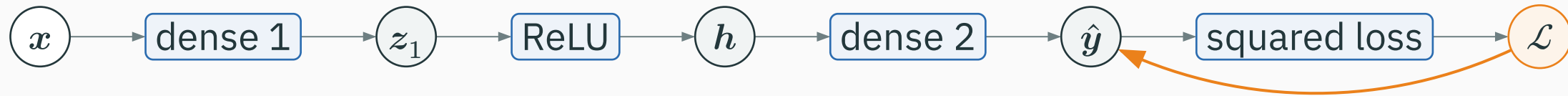


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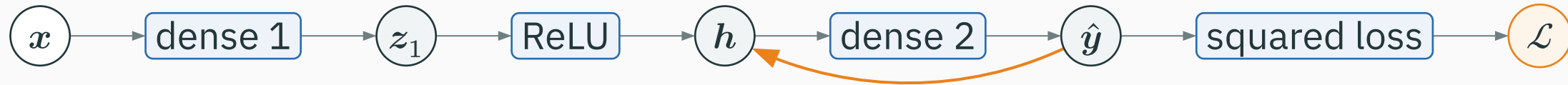
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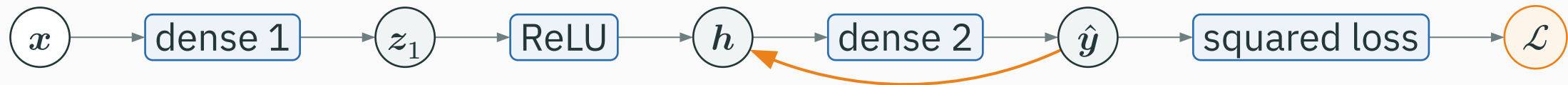
There is one returned number for each prediction coordinate: positive means increasing that prediction increases the loss; negative means it decreases the loss.

$g_{\hat{y}}$ now arrives at the second dense layer

Backward, part 2: dense 2 $\rightarrow$ hidden activations

D DERIVATION

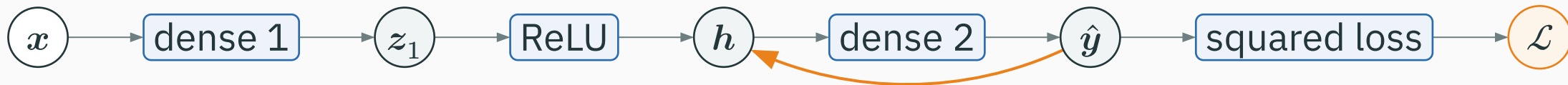




2. Second dense layer. It receives $\mathbf{g}_{\hat{y}} \in \mathbb{R}^{m \times 1}$ and returns a gradient to its input h :

$$\mathbf{g}_h = \mathbf{W}_2^\top \mathbf{g}_{\hat{y}}.$$

$$(h \times 1) = (h \times m)(m \times 1)$$



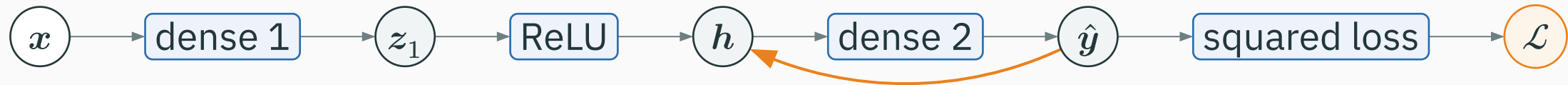
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It also returns parameter gradients

$$\partial \mathcal{L} / \partial \mathbf{W}_2 = \mathbf{g}_{\hat{y}} \mathbf{h}^\top \in \mathbb{R}^{m \times h}, \quad \partial \mathcal{L} / \partial \mathbf{b}_2 = \mathbf{g}_{\hat{y}} \in \mathbb{R}^{m \times 1}.$$



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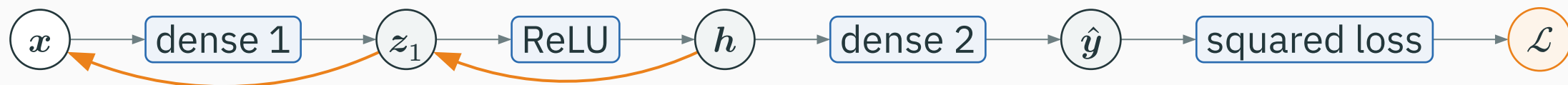
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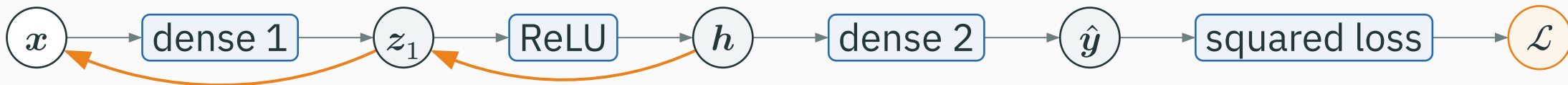
$$\partial \mathcal{L} / \partial \mathbf{W}_2 = \mathbf{g}_{\hat{y}} \mathbf{h}^\top \in \mathbb{R}^{m \times h}, \quad \partial \mathcal{L} / \partial \mathbf{b}_2 = \mathbf{g}_{\hat{y}} \in \mathbb{R}^{m \times 1}.$$

the returned $\mathbf{g}_h$ is now the arriving gradient at ReLU

Backward, part 3: ReLU $\rightarrow$ dense 1 $\rightarrow$ input

D DERIVATION

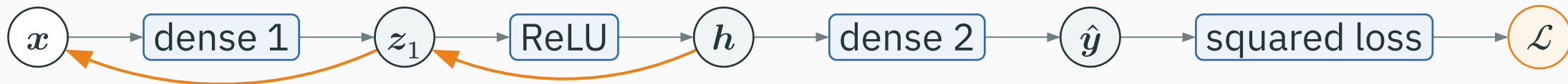




3. ReLU. Apply its local gate coordinate by coordinate:

$$g_{z_1} = g_h \odot \mathbb{1}[z_1 > 0].$$

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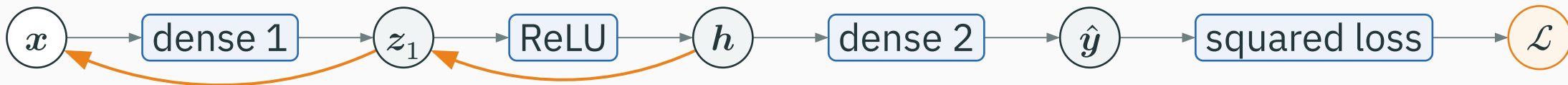
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4. First dense layer. It receives g_{z_1} and returns

$$g_x = W_1^\top g_{z_1}, \quad \partial \mathcal{L} / \partial W_1 = g_{z_1} x^\top, \quad \partial \mathcal{L} / \partial b_1 = g_{z_1}.$$

$$g_x : d \times 1 \quad \cdot \quad \partial \mathcal{L} / \partial W_1 : h \times d \quad \cdot \quad \partial \mathcal{L} / \partial b_1 : h \times 1$$



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$$g_x : d \times 1 \quad \cdot \quad \partial \mathcal{L} / \partial W_1 : h \times d \quad \cdot \quad \partial \mathcal{L} / \partial b_1 : h \times 1$$

each returned gradient becomes the arriving gradient for the block immediately to its left

The same tiny MLP in PyTorch

```
import torch
from torch import nn

model = nn.Sequential(
    nn.Linear(2, 3), nn.ReLU(), nn.Linear(3, 2)
)
x = torch.tensor([2., -1.])
y = torch.tensor([1., 0.])

y_hat = model(x)
loss = ((y_hat - y)**2).sum()
loss.backward()

for name, p in model.named_parameters():
    print(name, tuple(p.shape), tuple(p.grad.shape))
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PyTorch records the three blocks and applies their local rules in reverse

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/autograd

Step through the forward values, then follow one arriving gradient through a dense operation, a ReLU, and the preceding dense operation.

INTERACTIVE

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Step through the forward values, then follow one arriving gradient through a dense operation, a ReLU, and the preceding dense operation.

For practice, predict each gradient's shape before revealing its value in the interactive.

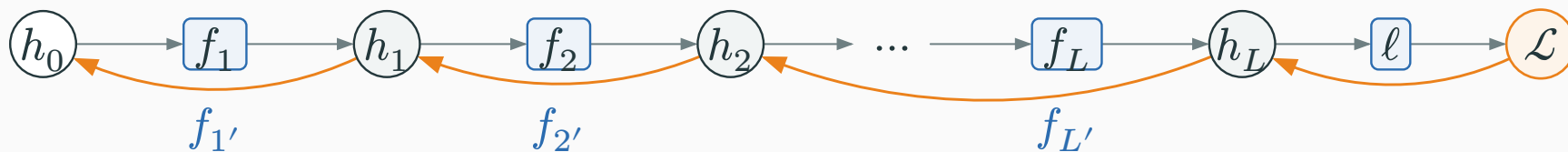
When gradients cross many layers

Deep networks repeat the same multiplication

For a scalar chain $h_0 \rightarrow h_1 \rightarrow \dots \rightarrow h_L \rightarrow \mathcal{L}$:

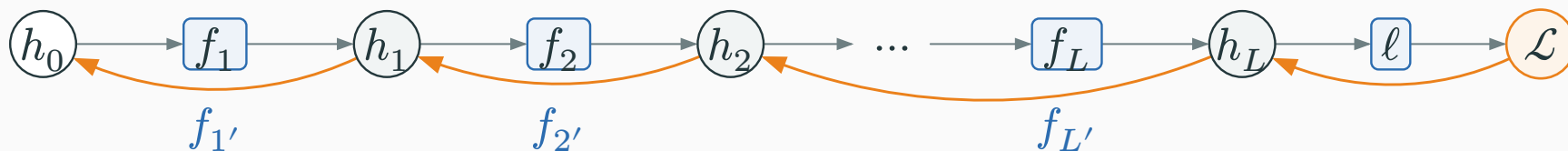
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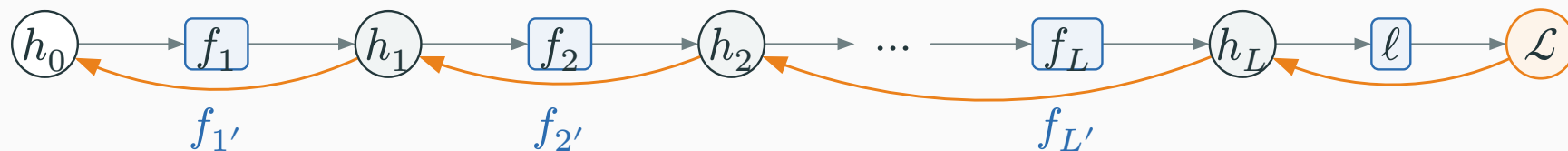
For a scalar chain $h_0 \rightarrow h_1 \rightarrow \dots \rightarrow h_L \rightarrow \mathcal{L}$:



$$\frac{\partial \mathcal{L}}{\partial h_0} = \frac{\partial \mathcal{L}}{\partial h_L} \cdot f_L'(h_{L-1}) \dots f_1'(h_0).$$

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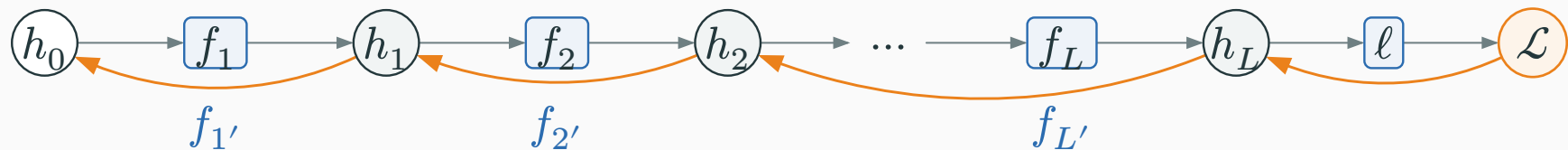


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depth creates a product of many local slopes

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depth creates a product of many local slopes

Vector layers do the same operation with matrix–vector products. The intuition is unchanged: repeatedly small factors shrink a gradient; repeatedly large factors grow it.

In a vector layer, the returned shape matches the previous value

D DERIVATION

For layer ℓ ,

$$z_\ell = W_\ell h_{\ell-1} + b_\ell.$$

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g_{z_ℓ}	$n_\ell \times 1$	arrives from the layer to the right
$W_\ell^\top$	$n_{\ell-1} \times n_\ell$	local linear map
$g_{h_{\ell-1}}$	$n_{\ell-1} \times 1$	returns to the previous layer

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each layer receives a gradient shaped like its output and returns one shaped like its input

Ten ordinary-looking slopes can create three regimes

Start with an arriving loss derivative of 1 and multiply the same local slope ten times:

Ten ordinary-looking slopes can create three regimes

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Local slope	After 10 layers	Effect
0.5	$0.5^{10} \approx 0.001$	gradient nearly disappears
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1.5	$1.5^{10} \approx 57.7$	gradient grows rapidly

Ten ordinary-looking slopes can create three regimes

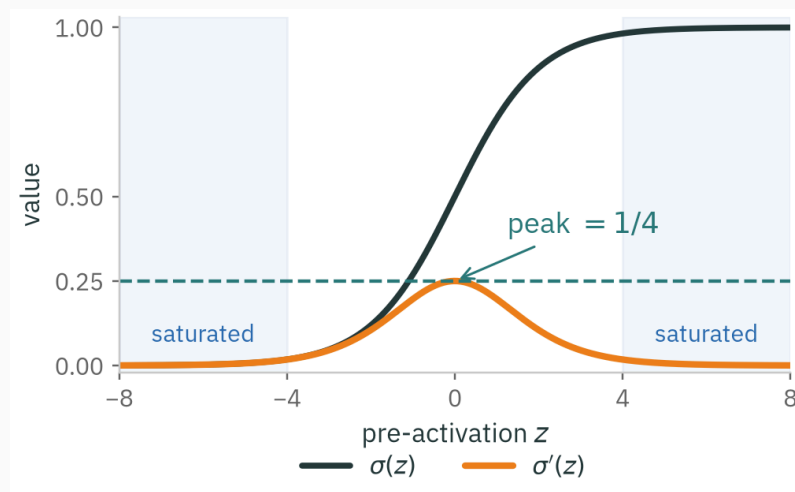
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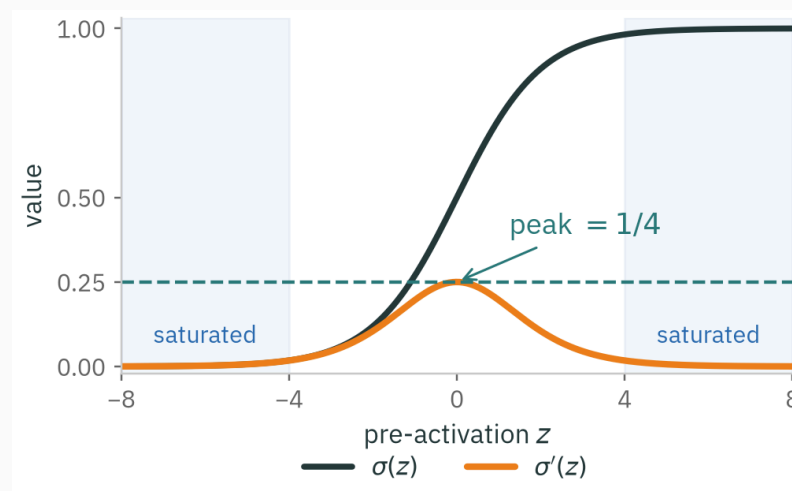
the chain rule explains both vanishing and exploding gradients

$\sigma'(z) = \sigma(z)(1 - \sigma(z)) \leq \frac{1}{4}$, and ≈ 0 for large $|z|$:

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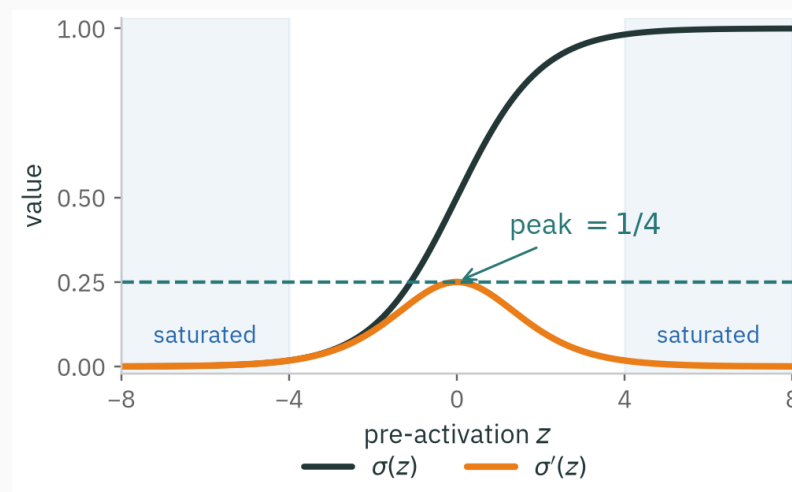


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Repeated sigmoid derivatives can shrink the signal quickly. ReLU has slope 1 on active units and 0 on inactive units; weights and normalization also shape the full Jacobian.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/vanishing-gradients

Sweep depth and activation function; watch the gradient magnitude at layer 0 collapse for sigmoid and survive for ReLU.

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Sweep depth to see repeated small local slopes rapidly erase an early-layer gradient.

Summary

Backprop in one sentence

Seed $\partial \mathcal{L} / \partial \mathcal{L} = 1$, then walk the graph **backward**, at each operation computing

downstream gradient = **upstream gradient** $\times$ **local derivative**,

and **adding** gradients wherever a variable branched.

Backprop in one sentence

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For vector operations, the **local derivative** can be a vector or matrix; the same color-coded chain rule applies.

Setting	Backward rule
scalar operation	$\frac{\partial \mathcal{L}}{\partial u} = \frac{\partial \mathcal{L}}{\partial v} \cdot \partial v / \partial u$
dense layer	$\frac{\partial \mathcal{L}}{\partial \mathbf{W}} = \frac{\partial \mathcal{L}}{\partial \mathbf{z}} \mathbf{x}^\top \frac{\partial \mathcal{L}}{\partial \mathbf{x}} = \mathbf{W}^\top \frac{\partial \mathcal{L}}{\partial \mathbf{z}}$
activation	$\frac{\partial \mathcal{L}}{\partial \mathbf{z}} = \frac{\partial \mathcal{L}}{\partial \mathbf{a}} \odot \varphi'(\mathbf{z})$
branch point	gradients add

We can compute $\partial \mathcal{L} / \partial \theta_j$ for every parameter.

Next: turn those gradients into reliable parameter updates.

Lecture 5 · Optimization for Deep Learning